

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.2	B1	0
Total 1 mark			

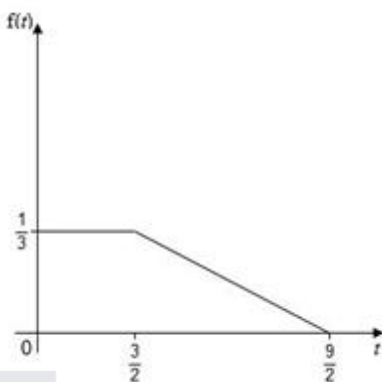
Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	States the mode	AO1.2	B1	1.75
(b)	Uses the area under a p.d.f. is equal to 1 to form an equation to find the y-coordinate of the maximum point of the p.d.f.	AO3.1a	M1	$\frac{2k}{2} = 1$ $k = 1$ Maximum point of the p.d.f. is (1.75, 1)
	Finds the equation of one line using their k , the y-coordinate of the maximum point of the p.d.f. May be in terms of k	AO1.1a	M1	$y = \frac{k}{1.75}x$ $y = \frac{1}{1.75}x$
	Finds the equation of both lines Follow through their k May be in terms of k	AO1.1b	A1F	$y = \frac{4}{7}x$
	States fully defined probability density function (OE) CAO	AO3.2a	A1	$y = -\frac{k}{0.25}(x-2)$ $y = -\frac{1}{0.25}(x-2)$ $y = -4x + 8$ $f(x) = \begin{cases} \frac{4}{7}x & 0 \leq x < 1.75 \\ -4x + 8 & 1.75 \leq x < 2 \\ 0 & \text{otherwise} \end{cases}$
				Total 5 marks

Q3.

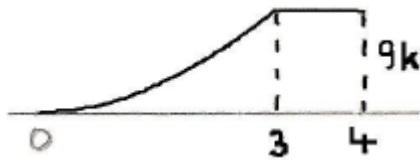
Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	4

Q4.

	Marking Instructions	AO	Marks	Typical Solution
(a)	(i) Draws horizontal line $(0, \frac{1}{3})$ to $(\frac{3}{2}, \frac{1}{3})$ and straight line joining $(\frac{3}{2}, \frac{1}{3})$ to t axis	AO1.1a	M1	
	Sketches correct shape, accurate and fully labelled	AO1.1b	A1	
(ii)	Deduces median is $\frac{3}{2}$	AO2.2a	B1	Area under $f(t)$ for $0 \leq t \leq \frac{3}{2}$ is $\frac{1}{2}$ Median of t is $\frac{3}{2}$
(b)	(i) States integral required for $E(T)$	AO1.1a	M1	$E(T) = \int_0^{\frac{3}{2}} \frac{1}{3} t dt + \int_{\frac{3}{2}}^{\frac{9}{2}} \frac{t(9-2t)}{18} dt$
	Obtains $E(T)$ correctly	AO1.1b	A1	$= \left[\frac{1}{6} t^2 \right]_0^{\frac{3}{2}} + \left[\frac{t^2}{4} - \frac{t^3}{27} \right]_{\frac{3}{2}}^{\frac{9}{2}} = \frac{13}{8}$
(ii)	Uses definition for variance (PI)	AO1.1a	M1	$\text{Var}(T) = E(T^2) - (E(T))^2$
	Obtains $\text{Var}(T)$	AO1.1b	A1	$\text{Var}(T) = \frac{15}{4} - \left(\frac{13}{8} \right)^2 = \frac{71}{64}$
	Obtains $\text{Var}(4T - 5)$ (Follow through 'their' $\text{Var}(T)$)	AO1.1b	A1F	$\text{Var}(4T - 5) = 16 \text{Var}(T) = \frac{71}{4}$
				Total 8 marks

Q5.

(a)



Curve + rectangle

B1

Some indication of x values

B1

$9k$ or 0.5 indicated for vertical height

B1

3

(b) Attempt to integrate kx^2 between 0 & 3

Show $\frac{kx^3}{3}$

M1

Obtain $9k$

A1

Area under rectangle = $9k$

B1

$9k + 9k = 1$ so $k = \frac{1}{18}$

For $9k + 9k = 1$. AG

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B1

4

(c) (i) 3

B1

(ii) Attempt to integrate kx^2 between 0 & Q_1 put = 0.25

M1

$\frac{(Q_1)^3}{54} = 0.25$

A1

$Q_1 = 2.38$

AWRT or any equivalent exact

form $\sqrt[3]{13.5} = \sqrt[3]{\frac{27}{2}} = \frac{3}{\sqrt[3]{2}} = \frac{3}{2}(\sqrt[3]{4})$

A1

4

[11]

Q6.

- (a) Putting $\frac{t^3}{216} = 0.9$

M1

$$t = 5.793$$

5.79 to 5.80

A1

41 days.

Accept 40 days in this context

A1

3

- (b) Attempt to differentiate $F(t)$

ct^2 seen

M1

$$f(t) = \frac{1}{72}t^2 \quad 0 \leq t \leq 6$$

Condone domain missing here

A1

$$= 0 \quad \text{otherwise}$$

For **complete** function

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A1

3

- (c) Attempt to integrate $tf(t)$ from 0 to 6

Using their $f(t)$ from **(b)** ct^4 seen

M1

$$E(T) = 4.5$$

A1

Attempt to integrate $t^2f(t)$ from 0 to 6

Using their $f(t)$ from **(b)** ct^5 seen

M1

$$E(T^2) = 21.6$$

A1

$$\text{Var}(T) = E(T^2) - E(T)^2$$

Applied in this case. Dependent on both M1

m1

$$= 21.6 - 4.5^2 = 1.35$$

A1

6

(d) S.d. = $\sqrt{1.35} = 1.162$

For $\sqrt{\text{their Var}} \ 0 < \text{Var}(T) < 9$

M1

Use of F(5.662)

For F(their s.d. + their E(T)) provided $0 < \text{Total} < 6$

m1

$$1 - \frac{5.662^3}{216}$$

m1

$$= 0.160$$

AWFW 0.159 to 0.161

A1

4

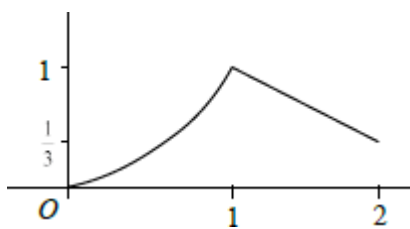
[16]

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Q7.

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(a)



Curve concave upwards between $(0, 0)$ and $(1, y_1)$

B1

Negative gradient line between $(1, y_1)$ and $(2, y_2)$ with $y_2 > 0$ (and not beyond 2)

B1

$y_1 = 1$ and $y_2 = \frac{1}{3}$ shown

B1

3

- (b) (i) Attempt to integrate t^2 between 0 and x

Accept integral of x^2

M1

$$F(x) = \frac{1}{3} x^3$$

A1

2

- (ii) Their $F(x) = 0.25$

M1

$$x = 0.909$$

AWRT; accept $\sqrt[3]{0.75}$ OE

A1

2

- (c) (i) $F(1) = \frac{1}{3}$

$$\int_1^x \frac{1}{3}(5-2t) dt = \left[\frac{1}{3}(5t-t^2) \right]_1^x$$

For integral attempted with correct limits

B1

M1

$$= \frac{1}{3}(5x-x^2) - \frac{4}{3}$$

For limits substituted in correct expression

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A1

$$F(x) = \frac{1}{3}(5x-x^2) - \frac{4}{3} + \frac{1}{3}$$

$F(1)$ added to give complete $F(x)$

A1

4

$$= \frac{1}{3}(5x-x^2-3)$$

AG

- (ii) $\frac{1}{3}(5q-q^2-3) = 0.75$

Setting up equation

or

integral of $f(x)$ from q to 2 = 0.25

M1

$$4q^2 - 20q + 21 = 0$$

or $q^2 - 5q + 5.25 = 0$

Reaching correct simplified quadratic

A1

$$(2q - 3)(2q - 7) = 0$$

or $q = 2.5 \pm 1$

Factorising for two solutions or using formula or calculator

m1

$$q = 1.5$$

Selecting only this one

A1

4

[15]

Q8.

(a)



Straight line from $(1, 0.5)$ to $(3, \frac{1}{6})$.

Horizontal straight line from $(3, \frac{1}{6})$ to $(5, \frac{1}{6})$.

B2,1

2

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(b)

$$E(X) = \frac{1}{6} \int_1^3 x(4-x) dx + \frac{1}{6} \int_3^5 x dx$$

ignore limits (**both** parts attempted)

M1

$$= \frac{1}{6} \left[2x^2 - \frac{x^3}{3} \right]_1^3 + \frac{1}{6} \left[\frac{x^2}{2} \right]_3^5$$

ignore limits (**both** correct)

A1

$$= \frac{1}{6} \left[(18-9) - \left(2 - \frac{1}{3} \right) \right] + \frac{1}{6} \left[\frac{25}{2} - \frac{9}{2} \right] = \frac{1}{6} \left[7\frac{1}{3} + 8 \right]$$

use of correct limits. dep on M1A1

m1

$$= 2\frac{5}{9}$$

(AG)

A1

4

(c) (i) $P(X > 2.5) = \frac{1}{3} + \frac{1}{2} \times \left(0.25 + \frac{1}{6}\right) \times \frac{1}{2}$

Or $1 - \int_1^{2.5} \frac{1}{6}(4-x) dx = 1 - \left[\frac{1}{6} \left(4x - \frac{x^2}{2} \right) \right]_1^{2.5}$

M1

$$= \frac{7}{16}$$

cao (0.4375)

A1

2

(ii) $P(1.5 < X < 4.5) = \frac{1}{2} \times \left(\frac{5}{12} + \frac{1}{6} \right) \times 1.5 \left\{ \begin{array}{l} + (4.5 - 3) \times \frac{1}{6} \end{array} \right\}$

Or $\int_{1.5}^3 \frac{1}{6}(4-x) dx + \int_3^{4.5} \frac{1}{6} dx$

M1

$$= \frac{7}{16} + \frac{1}{4}$$

A1

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$$= \frac{11}{16}$$

cao (= $\frac{11}{16}$ or 0.6875)

A1

3

(iii) $P(X > 2.5 \text{ and } 1.5 < X < 4.5) = P(2.5 < X < 4.5)$

$$\int_{2.5}^3 \frac{1}{6}(4-x) dx = \left[\frac{1}{6} \left(4x - \frac{x^2}{2} \right) \right]_{2.5}^3 = \frac{5}{48}$$

$$= \frac{1}{2} \times \left(0.25 + \frac{1}{6} \right) \times 0.5 + \frac{1}{4}$$

M1

$$= \frac{5}{48} + \frac{1}{4}$$

$$= \frac{17}{48}$$

cao (0.3541 $\dot{6}$)

A1

2

$$(iv) \quad P(X > 2.5 \mid 1.5 < X < 4.5) = \frac{\frac{17}{48}}{\frac{11}{16}}$$

their $\frac{(iii)}{(ii)}$ iff $0 < p's < 1$

M1

$$= \frac{17}{33}$$

cao (allow 0.51 $\dot{1}$)

A1

2

Alternative Solution

$$F(x) = \begin{cases} 0 & x < 1 \\ \frac{1}{12}(x-1)(7-x) & 1 \leq x < 3 \\ \frac{1}{6}(x+1) & 3 \leq x < 5 \\ 1 & x \geq 5 \end{cases}$$

$$(i) \quad P(X > 2.5) = 1 - F(2.5)$$

$$= 1 - \frac{1}{12} (2.5 - 1)(7 - 2.5)$$

©2025 Exam Papers Practice. All rights reserved. (M1)

$$= 1 - \frac{1}{12} \times 1.5 \times 4.5$$

$$= 1 - 0.5625$$

$$= 0.4375 \text{ or } \frac{7}{16}$$

cao

(A1)

(2)

$$(ii) \quad P(1.5 < X < 4.5) = F(4.5) - F(1.5)$$

$$= \frac{1}{6}(4.5+1) - \frac{1}{12} (1.5-1)(7-1.5)$$

(M1)

$$= \frac{11}{12} - \frac{11}{48}$$

(A1)

$$= \frac{11}{16} \text{ or } 0.6875$$

cao

(A1)

(3)

(iii) $P(X > 2.5 \text{ and } 1.5 < X < 4.5)$

$$= P(2.5 < X < 4.5)$$

$$= F(4.5) - F(2.5)$$

$$= \frac{11}{12} - \frac{9}{16}$$

(M1)

$$= \frac{17}{48}$$

cao

(A1)

(2)

(iv) $P(X > 2.5 \mid 1.5 < X < 4.5)$

(M1)

$$= \frac{F(4.5) - F(2.5)}{F(4.5) - F(1.5)} \text{ or } \frac{\text{their (iii)}}{\text{their (ii)}}$$

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$$= \frac{17/48}{11/12}$$

$$= \frac{17}{33} \text{ or (allow } 0.51)$$

cao

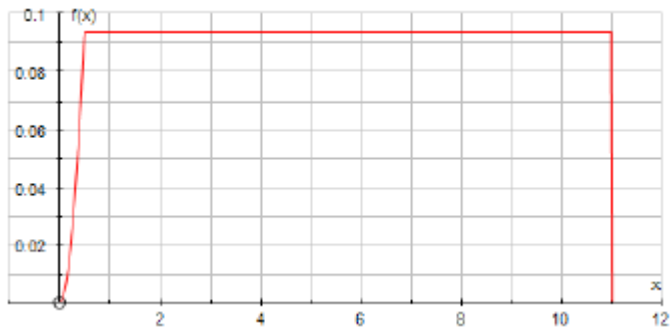
(A1)

(2)

[15]

Q9.

(a)



B1 for concave curve from

(0, 1) to $\left(\frac{1}{2}, \frac{3}{32}\right)$

B1 for horizontal straight line

$f = \frac{3}{32}$ from $\left(\frac{1}{2}, \frac{3}{32}\right)$ to $\left(\frac{11}{2}, \frac{3}{32}\right)$

B1 for correct axes

B3

3

(b) (i)
$$P\left(X \geq 8\frac{1}{3}\right) = \left[\frac{3}{32} \times \left(11 - 8\frac{1}{3}\right)\right]$$

$$= \frac{3}{32} \times \frac{8}{3}$$

Any correct method attempted in either part

M1

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AG

A1

(ii)
$$P(X \geq 3) = \frac{3}{32} \times (11 - 3)$$

$$= \frac{3}{4}$$

Any correct method attempted

AG

A1

3

(c) (i) Interquartile Range = $5\frac{1}{3}$
cao

B1

(ii) Median = $5\frac{2}{3}$
Cao

B2

Alternative:

$$\frac{1}{64} + \frac{3}{32} \left(m - \frac{1}{2} \right) = \frac{1}{2}$$

$$\Rightarrow 3 \left(m - \frac{1}{2} \right) = 15.5 \Rightarrow m = 5\frac{2}{3}$$

sc if B0 then:
M1 for correct method seen

$$\frac{1}{2} \left(8\frac{1}{3} + 3 \right) \text{ or } \frac{1}{2} \times 11\frac{1}{3}$$

$$\text{or } \frac{3}{32} (11 - m) = \frac{1}{2} \Rightarrow 11 - 5\frac{1}{3}$$

3

(d) $P[(X < m) \cap (X \geq 3)] = \frac{1}{4}$
 $\left(\frac{3}{4} - \frac{1}{2} \right)$ attempted

B1

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$$P(X < m | X \geq 3) = \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}$$

(their p) $\frac{1/4}{3/4}$ for $0 < p < 1$

M1

cao

A1

Alternative:

(Ratio of relevant two areas)

$$P(X < m | X \geq 3) = \frac{2\frac{2}{3}}{8} = \frac{1}{3}$$

cao

3

[12]

Q10.

(a) (i) $E(T) = \frac{1}{2} (25 + -5) = 10$
 CAO

B1

1

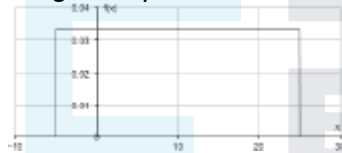
(ii) $Var(T) = \frac{1}{12} (25 - -5)^2 = 75$
 CAO

B1

1

(b) $P(-2 < T < 2) = \frac{2}{15}$ (OE)

Diagram optional



B1

P(magnitude at least 2 minutes)

$= 1 - P(-2 < T < 2)$

$= 1 - \frac{4}{30}$

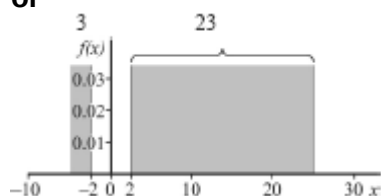
M1

$= \frac{13}{15}$ (OE) = 0.867

CAO (AWRT)

A1

or



$\frac{1}{30} (3 + 23) = \frac{26}{30} = \frac{13}{15}$

Alternative

$P(T > 2) = \frac{23}{30} (0.766\dot{6})$

$$\text{or } P(T < -2) = \frac{1}{10}$$

B1

or

$$\int_{-5}^{-2} \frac{1}{30} dt + \int_2^{25} \frac{1}{30} dt = \frac{1}{10} + \frac{23}{30} = \frac{13}{15}$$

$$\begin{aligned} &P(\text{magnitude at least 2 minutes}) \\ &= P(T < -2) + P(T > 2) \\ &= \frac{13}{15} \text{ for M1 A1} \end{aligned}$$

or

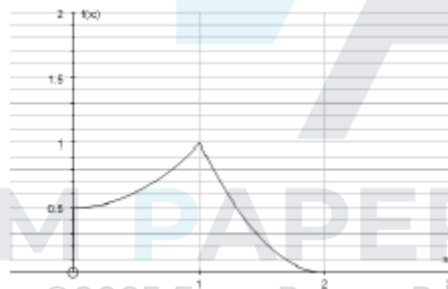
$$\begin{aligned} 1 - \int_{-2}^2 \frac{1}{30} dt &= 1 - \left[\frac{t}{30} \right]_{-2}^2 \\ &= 1 - \frac{4}{30} = \frac{26}{30} = \frac{13}{15} \end{aligned}$$

3

[5]

Q11.

(a)



B1 for axes

B1 for curve from (0, 0.5) to (1, 1)

B1 for curve from (1, 1) to (2, 0)

B3

3

(b)

$$P(X \leq 1) = \int_0^1 \frac{1}{2}(x^2 + 1) dx$$

M1

$$= \left[\frac{x^3}{6} + \frac{x}{2} \right]_0^1$$

A1

$$= \left[\frac{1}{6} + \frac{1}{2} \right] = \frac{2}{3}$$

0.667

A1

3

(c) $E(X^2) = \int_0^1 x^2 \times \frac{1}{2}(x^2 + 1) dx + \int_1^2 x^2(x - 2)^2 dx$

both integrals seen

M1

$$= \left[\frac{x^5}{10} + \frac{x^3}{6} \right]_{x=0}^{x=1} + \left[\frac{x^5}{5} - x^4 + \frac{4x^3}{3} \right]_{x=1}^{x=2}$$

A1A1

$$= \left(\frac{1}{10} + \frac{1}{6} \right) + \left(\left[\frac{32}{5} - 16 + \frac{32}{3} \right] - \left[\frac{1}{5} - 1 + \frac{4}{3} \right] \right)$$

dep(M1)

m1

$$= \frac{4}{5}$$

AG

A1

5

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(d) (i) $E(X) = \frac{19}{24}$ and $k\text{Var}(X) = 499$

$$\text{Var}(X) = E(X^2) - E^2(X)$$

$$= \frac{4}{5} - \left(\frac{19}{24} \right)^2$$

M1

$$= \frac{499}{2880} \quad (0.173)$$

A1

$$\Rightarrow k = 2880$$

CAO

A1

3

$$\begin{aligned}
 \text{(ii)} \quad E(5X^2 + 24X - 3) \\
 = 5E(X^2) + 24E(X) - 3
 \end{aligned}$$

M1

$$= 5 \times \frac{4}{5} + 24 \times \frac{19}{24} - 3$$

$$= 20$$

CAO

A1

2

$$\text{(iii)} \quad \text{Var}(12X - 5) = 144\text{Var}(X)$$

M1

$$= 144 \times \frac{499}{2880}$$

$$= \frac{499}{20} \text{ or } (24.95)$$

CAO (AWFW 24.9 to 25)

A1

2

[18]

Q12.

$$\text{(a)} \quad \text{Median} = 1$$

B1

$$\text{Lower quartile} = \frac{1}{2}$$

B1

2

$$\begin{aligned}
 \text{(b)} \quad F(1) &= \frac{1}{2} \\
 \text{For } 1 \leq x \leq 4 \\
 \int \frac{1}{18} (x-4)^2 dx
 \end{aligned}$$

ignore limits

M1

$$\begin{aligned}
 &= \left[\frac{1}{54} (x-4)^3 \right]_1^x \\
 &= \left[\frac{1}{54} (x-4)^3 + \frac{1}{2} \right]
 \end{aligned}$$



A1

$$F(x) = \left[\frac{1}{54} (x-4)^3 + \frac{1}{2} \right] + \frac{1}{2}$$

adding $\frac{1}{2}$ **or** $F(1)$

m1

$$= 1 + \frac{1}{54} (x-4)^3$$

CAO (AG)

A1

Alternative

$$\int \frac{1}{18} (x-4)^2 dx = \frac{1}{54} (x-4)^3 + c$$

(M1)

$$F(1) = \frac{1}{2} \Rightarrow c = 1$$

(m1)(A1)

$$F(x) = 1 + \frac{1}{54} (x-4)^3$$

(A1)

4

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Alternative

$$\int \frac{1}{18} (x-4)^2 dx \quad (M1)$$

$$= \int_1^x \frac{1}{18} (x^2 - 8x + 16) dx$$

$$= \frac{1}{18} \left[\frac{x^3}{3} - 4x^2 + 16x \right]_1^x \quad (A1)$$

$$F(x) = \frac{1}{2} + \frac{1}{54} [x^3 - 12x^2 + 48x]_1^x \quad (m1)$$

$$= \frac{1}{2} + \frac{1}{54} (x^3 - 12x^2 + 48x - 37)$$

$$= 1 + \frac{1}{54} (x^3 - 12x^2 + 48x - 64)$$

$$= 1 + \frac{1}{54} (x-4)^3 \quad (A1)$$

$$(c) \quad P(2 \leq X \leq 3) = \frac{53}{54} - \frac{46}{54}$$

$$F(3) - F(2)$$

M1

$$= \frac{7}{54} (0.130)$$

$$0.129\bar{6}$$

A1

2

$$(d) \quad (i) \quad F(q) = \frac{3}{4}$$

$$\text{use of } F(q) = \frac{3}{4}$$

M1

$$1 + \frac{1}{54} (q - 4)^3 = \frac{3}{4}$$

$$\frac{1}{54} (q - 4)^3 = -\frac{1}{4}$$

(either)

M1

$$(\times 54) \Rightarrow (q - 4)^3 = -13.5$$

AG

A1

3

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$$(ii) \quad q - 4 = \sqrt[3]{-13.5} = -2.3811 \quad q = 1.619 \text{ (3dp)}$$

CAO

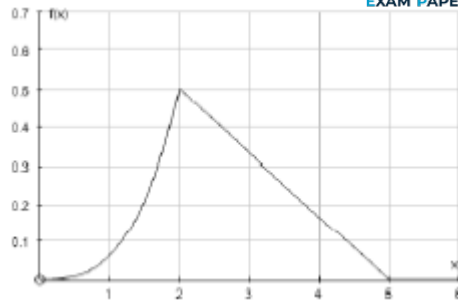
B1

1

[12]

Q13.

(a)



for concave curve from $(0, 0)$ to $(2, 0.5)$

B1

for straight line from $(2, 0.5)$ to $(5, 0)$

B1

for axes $[2, 5; 0.5]$ seen

B1

3

(b) $P(X \geq 2) = \frac{1}{2} \times 3 \times 0.5 = 0.75$
 $\Rightarrow F(2) = 0.25$

Alternative:

$$\int \frac{1}{6}(5-x)dx = \frac{1}{6} \times \frac{(5-x)^2 \times (-1)}{2}$$

$$= -\frac{1}{12}(5-x)^2$$

$$2 \leq x \leq 5$$

$$F(x) = F(2) + \int_2^x \frac{1}{6}(5-x)dx$$

$$= 0.25 + \frac{1}{6} \left[5x - \frac{x^2}{2} \right]_2^x$$

M1

$$= 0.25 + \frac{1}{6} \left(5x - \frac{x^2}{2} \right) - \frac{1}{6} (10 - 2)$$

A1

$$= 0.25 - \frac{8}{6} + \frac{5x}{6} - \frac{x^2}{12}$$

$$= -\frac{1}{12}(x^2 - 10x + 13)$$

M1

$$= 1 - \frac{1}{12}(5-x)^2$$

A1

Or

$$F(x) = 1 - \text{Area } \Delta (\text{base } x, 5)$$

$$= 1 - \frac{1}{2}(5-x)\frac{1}{6}(5-x)$$

$$= 1 - \frac{1}{12}(5-x)^2$$

4

(c) $P(X \leq 4) = F(4)$
 $= 1 - \frac{1}{12}(5-4)^2 = \frac{11}{12} \quad (0.916 \text{ to } 0.917)$

B1

$$F(3) = 1 - \frac{1}{12}(2)^2 = \frac{2}{3} \quad (0.667)$$

B1

$$P(X \geq 3 \text{ and } X \leq 4) = F(4) - F(3)$$

$$= \frac{11}{12} - \frac{2}{3} = \frac{1}{4} \quad (0.25)$$

B1

$$P(X \geq 3 | X \leq 4) = \frac{F(4) - F(3)}{F(4)}$$

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M1

$$= \frac{\frac{1}{4}}{\frac{11}{12}} = \frac{3}{11}$$

A1

Alternative:

$$P(X \geq 3 | X \leq 4)$$

$$= \frac{F(4) - F(3)}{F(4)} \quad (\text{M1})$$

$$= 1 - \frac{F(3)}{F(4)}$$

$$= 1 - \frac{\frac{2}{3}}{\frac{11}{12}}$$

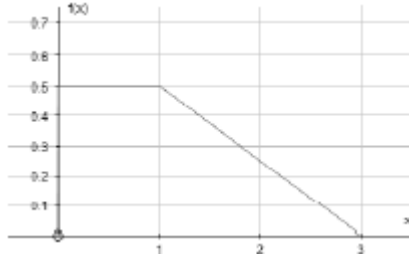
$$\begin{aligned}
 &= 1 - \frac{8}{11} (B1) (0.727\dot{2}) \\
 &= \frac{3}{11} \text{ (AWFW 0.272 to 0.273)}
 \end{aligned}$$

5

[12]

Q14.

(a) Sketch:



1 for straight line $0 \leq x \leq 1$ from $(0, 0.5)$ to $(1, 0.5)$

1 for straight line $1 \leq x \leq 3$ from $(1, 0.5)$ to $(3, 0)$

1 for axes [must have at least $(0, 0.5)$
 $(1, 0)$ and $(3, 0)$ labelled]

B3

3

(b) $P(X \leq \eta) = F(\eta) = 0.5$

M1

($\Rightarrow \eta = 1$ (from graph))

AG

A1

2

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(c)
$$\mu = E(X) = \int_0^1 \left(\frac{x}{2}\right) dx + \int_1^3 x \left(\frac{3-x}{4}\right) dx$$

Both integrals stated

M1

$$= \left[\frac{x^2}{4} \right]_0^1 + \frac{1}{4} \left[\frac{3x^2}{2} - \frac{x^3}{3} \right]_1^3$$

Either

A1

$$= \frac{1}{4} + \frac{1}{4} \left[\left(\frac{27}{2} - 9 \right) - \left(\frac{3}{2} - \frac{1}{3} \right) \right]$$

Correct limits used on both

m1

$$= \frac{1}{4} + \frac{5}{6} \quad (0.25 + 0.8\dot{3})$$

$$= 1\frac{1}{12}$$

(CAO)

A1

4

(d) Area of Δ

$$= P\left(X > 2\frac{1}{4}\right) = \frac{1}{2} \times \frac{3}{4} \times \frac{3 - 2\frac{1}{4}}{4}$$

M1ft

$$= \frac{3}{32} \times \frac{3}{4} = \frac{9}{128}$$

$$\therefore P\left(X < 2\frac{1}{4}\right) = 1 - \frac{9}{128}$$

M1ft

$$= \frac{119}{128} (0.9296875)$$

A1

Alternative:

For $1 \leq x \leq 3$

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$$F(x) = 1 - \frac{1}{8}(3-x)^2 \quad M1ft$$

$$F\left(2\frac{1}{4}\right) = 1 - \frac{1}{8} \times \frac{9}{16} \quad M1ft$$

$$= \frac{119}{128}$$

CAO

or

$$\int_{2\frac{1}{4}}^3 \frac{3-x}{4} dx \left(= \frac{9}{128} \right)$$

M1ft

$$= 1 - \int_{2\frac{1}{4}}^3 \frac{3-x}{4} dx$$

M1ft

$$\begin{aligned} &= 1 - \frac{1}{4} \left[3x - \frac{x^2}{2} \right]_{2\frac{1}{4}}^3 \\ &= 1 - \frac{1}{4} \left[9 - \frac{9}{2} - \frac{27}{4} + \frac{81}{32} \right] \\ &= 1 - \frac{1}{4} \times \frac{9}{32} = \frac{119}{128} \end{aligned}$$

A1

or $(1 - 0.0703125 = 0.9296875)$

Alternative

$$f\left(2\frac{1}{4}\right) = \frac{3}{16} = 0.1875$$

$$P(X < 3\mu - \eta) = P\left(X < 2\frac{1}{4}\right)$$

$$= \frac{1}{2} + \frac{1}{2} \left(\frac{3}{16} + \frac{1}{2} \right) \times 1\frac{1}{4} \quad \text{M1ft}$$

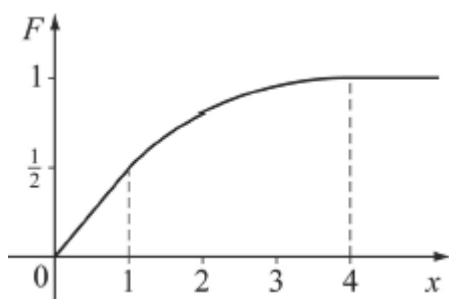
$$= \frac{1}{2} + \frac{55}{128} (0.4296875) \quad \text{M1ft}$$

$$= \frac{119}{128} (0.930) \quad \text{A1}$$

3
EXAM PAPERS PRACTICE [12]

Q15. ©2025 Exam Papers Practice. All rights reserved.

(a) (i)



B1 for axes 0 to 4 & 0 to 1

B1 for straight line $0 - \left(1, \frac{1}{2}\right)$

B1 for convex curve from $\left(1, \frac{1}{2}\right)$ to $(4,1)$

B1 for at least the straight line for $x > 4$

B4

4

$$(ii) \quad F(q_1) = 0.25 \Rightarrow \frac{1}{2}q_1 = 0.25$$

From sketch, or from $F(x)$, Median = 1.

M1

$$\Rightarrow q_1 = \frac{1}{2}$$

$\frac{1}{2}x$ is linear on $(0, 0)$ to $\left(1, \frac{1}{2}\right)$

$$\therefore q_1 = \frac{1}{2}$$

AG

A1

2

$$(iii) \quad F(1.6) = 0.744 \\ F(1.7) = 0.775$$

M1

$$F(q_3) = 0.75$$

M1

$$\Rightarrow 1.6 < q_3 < 1.7$$

AG

A1

3

$$(b) \quad (i) \quad f(x) = F(x)$$

M1

$$\Rightarrow f(x) = \frac{1}{2} \text{ for } 0 \leq x \leq 1$$

A1

$$\Rightarrow a = \frac{1}{2}$$

$$f(1) = a = 9\beta$$

B1

$$\int_{-\infty}^{\infty} f(x) dx = 1 \Rightarrow$$

$$[ax]_0^1 + \left[\frac{\beta(x-4)^3}{3} \right]_1^4 = 1 \quad \text{M1}$$

$\Rightarrow$ for $1 \leq x \leq 4$

$$f(x) = \frac{1}{54} (3x^2 - 24x + 48)$$

A1

$$= \frac{3}{54} (x^2 - 8x + 16)$$

M1

$$= \frac{1}{18} (x - 4)^2$$

$$\Rightarrow \beta = \frac{1}{18}$$

A1

$$\Rightarrow \alpha + 9\beta = 1$$

A1

Solving:

M1

$$\alpha = \frac{1}{2} \text{ and } \beta = \frac{1}{18}$$

A1

5

(ii) $E(X) = \int_0^1 \frac{1}{2} x dx + \int_1^4 \frac{1}{18} (x^3 - 8x^2 + 16x) dx$

Both seen

M1

$$= \left[\frac{1}{4} x^2 \right]_0^1 + \frac{1}{18} \left[\frac{x^4}{4} - \frac{8x^3}{3} + 8x^2 \right]_1^4$$

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A1A1

$$= \frac{1}{4} + \frac{7}{8}$$

Dependent on M1

m1

CAO

A1

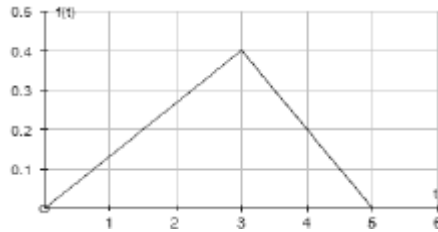
$$= 1\frac{1}{8}$$

5

[19]

Q16.

(a)



line segment on 0 – 3

B1

line segment on 3 – 5

B1

scales (0 – 0.4 vertical; 0 – 5 horizontal)

B1

3

(b) (i)

$$P(T \leq 2) = \frac{1}{2} \times 2 \times \frac{4}{15}$$

M1

$$= \frac{4}{15}$$

(0.267)

A1

2

(ii) $P(2 < T < 4) = 1 - (P(T < 2) + P(T > 4))$

M1

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$$= 1 - \left(\frac{4}{15} + \frac{1}{2} \times \frac{1}{5} \right)$$

$$\text{for } P(T > 4) = \frac{1}{10}$$

$$\frac{1}{2} d[(f_2 + f_4) + 2f_3]$$

$$f_2 = \frac{4}{15}; f_4 = \frac{1}{5}; f_3 = \frac{2}{5}; d = 1$$

A1

$$= 1 - \frac{4}{15} - \frac{1}{10}$$

$$= \frac{19}{30}$$

(0.633)

A1

3

(c)
$$E(T) = \int_0^3 \frac{2}{15} t^2 \, dt = \int_3^5 t \left(1 - \frac{1}{5} t\right) \, dt$$

both integrals seen

M1

$$= \left[\frac{2}{45} t^3 \right]_0^3 + \left[\frac{1}{2} t^2 - \frac{1}{15} t^3 \right]_3^5$$

B1B1

$$= \frac{6}{5} + \frac{25}{6} - \frac{27}{10}$$

$$= 2\frac{2}{3}$$

OE

A1

4

[12]

Q17.

(a) (i)
$$E\left(\frac{1}{X}\right) = \int_0^1 \frac{1}{x} 3x^2 \, dx = \int_0^1 3x \, dx$$

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M1

$$= \left[\frac{3x^2}{2} \right]_0^1 = 1.5$$

CAO

A1

A1

3

(ii)
$$E\left(\frac{1}{X^2}\right) = \int_0^1 \frac{1}{x^2} 3x^2 \, dx = \int_0^1 3 \, dx$$

M1

$$= [3x]_0^1 = 3.0$$

A1

$$\text{Var}\left(\frac{1}{X}\right) = 3.0 - (1.5)^2$$

dependent on previous M1

m1

$$= 0.75$$

[on their (i)] and Var > 0

A1ft

4

(b)
$$E\left(\frac{5+2X}{X}\right) = E\left(\frac{5}{X} + 2\right)$$

M1

$$= 5E\left(\frac{1}{X}\right) + 2$$

M1

$$= 5 \times 1.5 + 2$$

$$= 9.5$$

CAO

A1

$$\text{Var}\left(\frac{5+2X}{X}\right) = \text{Var}\left(\frac{5}{X} + 2\right)$$

$$= 25 \times \text{Var}\left(\frac{1}{X}\right)$$

M1

$$= 25 \times 0.75$$

$$= 18.75$$

CAO

A1

5

[12]

Q18.

(a)
$$E(T) = \int_0^1 t f(t) dt$$

$$= \int_0^1 4t^2 (1-t^2) dt$$

M1

$$= \left(\frac{4t^3}{3} - \frac{4t^5}{5} \right) \Big|_0^1$$

A1

$$= \frac{4}{3} - \frac{4}{5}$$

A1

$$= \frac{8}{15}$$

AG

3

(b) (i) $F(t) = P(T < t) = \int_0^t f(t) dt$

$$= \int_0^t 4t(1-t^2) dt$$

M1

$$= \left(2t^2 - t^4 \right) \Big|_0^t$$

$$= 2t^2 - t^4$$

A1

2

(ii) $P(\mu < T < m) = F(m) - F(\mu)$

$\Downarrow$

M1

$$F(m) = 0.5$$

B1

$$F(\mu) = F\left(\frac{8}{15}\right) = 0.4880$$

B1

$$P(\mu < T < m) = 0.5 - 0.488$$

0.5—their $F(\mu)$

M1ft

$$\therefore = 0.012$$

A1

5

[10]

Q19.

(a) $\alpha = \frac{1}{0.4 - (-0.4)} = 1.25$

B1

1

(b) $E(R) = \frac{1}{2}(-0.4 + 0.4) = 0$

B1

$$\text{Var}(R) = \frac{1}{12}(0.8)^2 = \frac{4}{75}$$

M1

Standard deviation $= \frac{2}{5\sqrt{3}} = 0.231$

A1

3

(c) $P(|R| < 3) = P(-0.3 < R < 0.3)$
(or seen on diagram)

M1

$$= 0.6 \times 1.25$$

$$= 0.75$$

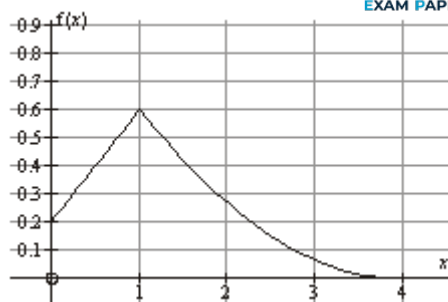
A1

2

[6]

Q20.

(a)



B1 for line segment

(0, 0.2) to (1, 0.6)

B1 for correct shaped curve

(1, 0.6) to (4, 0)

B2

2

(b) (i) for $0 \leq x \leq 1$

$$F(x) = \int_0^x \frac{1}{5} (2x) dx$$

M1

$$= \frac{1}{5} (x^2 + x) \Big|_0^x$$

A1

$$\frac{1}{5} x(x+1)$$

A1

3

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(ii) $P(X \geq 0.5) = 1 - F(0.5)$

$$= 1 - \frac{1}{5} \times 0.5 \times 1.5$$

M1

$$= 1 - 0.15$$

$$= 0.85$$

(OE)

A1

2

(iii) $F(q_1) = 0.25$

$$\frac{1}{5} x(x+1) = 0.25$$



B1

$$F(0.5) = \frac{3}{20} = 0.15 < 0.25$$

$$F(0.75) = \frac{21}{80} = 0.2625 > 0.25$$

Attempt to solve



$$q_1 = 0.725$$



M1

$$\therefore 0.5 < q_1 < 0.75$$

$$0.5 < q_1 < 0.75$$

AG

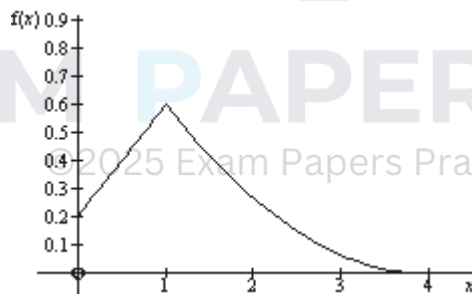
A1

3

[10]

Q21.

(a)



B1 for line segment (0,0.2) to (1,0.6)

B1 for correctly shaped curve (1,0.6) to (4,0)

B2

2

(b) (i) for $0 \leq x \leq 1$

$$F(x) = \int_0^x \frac{1}{5}(2x+1)dx$$

Ignore limits

M1

$$= \left[\frac{1}{5} (x^2 + x) \right]_0^x$$

Ignore limits

A1

$$= \frac{1}{5} x(x+1)$$

A1

3

(ii)

$$P(X \leq 1) = F(1)$$

$$= \frac{2}{5}$$

B1

1

(iii)

$$P(X \geq x) = \frac{17}{20} \Rightarrow F(x) = \frac{3}{20}$$

M1

$$\frac{1}{5} x(x+1) = \frac{3}{20}$$

m1

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$$x(x+1) = \frac{3}{4}$$

$$x^2 + x - \frac{3}{4} = 0$$

A1

$$\left(x - \frac{1}{2}\right)\left(x + \frac{3}{2}\right) = 0$$

Any valid method attempted

m1

$$x - \frac{1}{2}$$

A1

5

(iv) Since $F(1) = 0.4$, q lies in $0 \leq r \leq 1$

$$F(q) = \frac{1}{5} (q^2 + q) = 0.25$$

M1

$$\Rightarrow q^2 + q = 1.25$$

$$q^2 + q - 1.25 = 0$$

A1

$$\Rightarrow q = \frac{-1 \pm \sqrt{1 - 4 \times (-1.25)}}{2}$$

m1

$$q = \frac{1}{2} (\sqrt{6} - 1) \quad (q > 0)$$

AWFW (0.724 to 0.725)

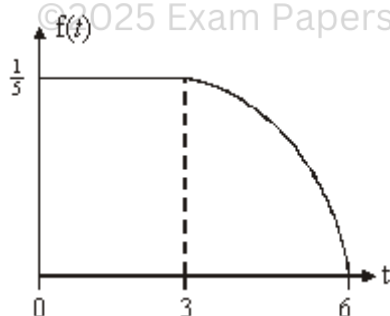
A1

4

[15]

Q22.

(a)



B1 2 axes with scales

B1 Horizontal line at 0.2 from 0 to 3

B1 Curve from 3 to 6

B3

3

(b) $P(T = 3) = 0$

B1

1

(c) $P(T \geq 3) = 1 - P(T < 3)$

M1

$$= 1 - \frac{3}{5}$$

$$= \frac{2}{5}$$

$$\int_3^6 \frac{1}{45} t(6-t) dt = \frac{2}{5}$$

A1

2

(d) $\int_0^{\infty} \frac{1}{5} dt = 0.5$

$$P(T \leq 3) = 0.6$$

M1

$$\therefore \left(\frac{t}{5}\right)_0^{\infty} = 0.5$$

$$\therefore 0 \leq \text{median} < 3$$

$$\frac{m}{5} - 0 = 0.5$$

$$\frac{1}{5} m = 0.5$$

$$m = 0.5 \times 5$$

$$m = 5 \times 0.5$$

$$m = 2.5$$

$$m = 2.5 \quad \text{AG}$$

A1

2

(e) $E(T) = \int_0^3 \frac{1}{5} t dt + \int_3^6 \frac{1}{45} t^2 (6-t) dt$

M1

$$= \left[\frac{1}{10} t^2 \right]_0^3 + \left[\frac{2}{45} t^3 - \frac{1}{180} t^4 \right]_3^6$$

A1A1

$$= \frac{9}{10} + 1.65$$

$$= 2.55$$

A1

$$\therefore P(\text{median} < T < \text{mean})$$

$$= P(2.5 < T < 2.55)$$

M1

$$= 0.05 \times \frac{1}{5}$$

$$= 0.01$$

A1

6

[14]

Q23.

(a) $k = 0.1$

OE.

B1

1

(b) $E(X) = 1$

B1

1

(c) $P(X > 0) = 6 \times 0.1$

M1

$$= 0.6$$

A1

2

(d) $P(|X| > 3.5) = 1 - P(|X| < 3.5)$

M1

$$= 1 - 0.7$$

A1

$$= 0.3$$

A1

3

Alternative solution
 $P(X < -3.5) + P(X > 3.5)$

(M1)

$$= \frac{0.5}{10} + \frac{2.5}{10}$$

(A1)

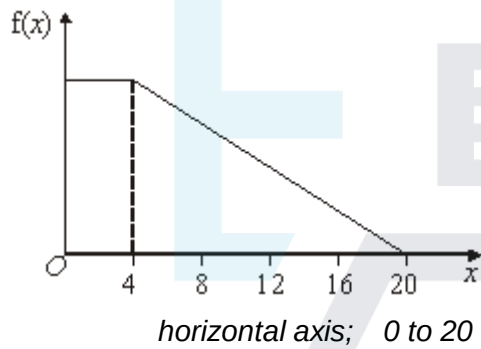
$$= \frac{3}{10}$$

(A1)

[7]

Q24.

(a)



B1

vertical axis; 0 to c or 1/12

B1

horizontal line @ c from 0 to 4

B1

Line from (4,c) to (20,0)

B1

4

(b) Area under graph = 1
 use of

M1

Area under graph =

$$4c + \frac{1}{2}(20 - 4)c = 12c$$

area of (rectangle + triangle)

M1

or

$$= \frac{c}{2} (4 + 20) = 12c$$

area of trapezium

Hence $12c = 1$ so $c = \frac{1}{12}$

*CAO; not decimal equivalent
(but accept 0.083)*

A1

3

(c) $P(\text{Length} < 2.01) = P(X < 10)$

CAO

B1

$$f(10) = \frac{5c}{8} = \frac{5}{96} = 0.0521$$

CAO/AWRT; ft on c only

B1ft

$$P(X < 10) = 4c + \frac{1}{2} \left(c + \frac{5c}{8} \right) 6$$

area of (rectangle + trapezium)

or $\int_2^4 c dx + \int_4^{10} \frac{c}{16} (20 - x) dx$ etc

1 - area of (triangle)

M1

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or

$$= 1 - \frac{1}{2} (20 - 10) \frac{5c}{8}$$

$$= \frac{71c}{8} \text{ or } 1 - \frac{25c}{8}$$

$$[cx] + \left[\frac{c}{16} \left(20x - \frac{x^2}{2} \right) \right]_4^{10} \quad \text{A1}$$

$$= \frac{71}{96} \text{ or } 0.739 \text{ to } 0.740$$

CAO/AWRT; accept 0.74

A1

4

[11]

Q25.

(a) $P(2 \leq X \leq 3) = F(3) - F(2)$

M1

$$= \frac{27}{64} - \frac{1}{8}$$

$$= \frac{19}{64} = 0.297$$

CAO/AWRT

A1

2

(b) $f(x) = \begin{cases} \frac{3x^2}{64} \\ 0 \leq x \leq 4 \end{cases}$

M1A1

2

(c) (i) $E(X) = \int_0^4 x \cdot f(x) dx$

$$= \int_0^4 \frac{3x^3}{64} dx$$

M1

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A1ft

$$= 3$$

CAO

A1

3

(ii) $\text{Var}(X) = \int x^2 f(x) dx - [E(X)]^2$

$$= \int_0^4 \frac{3x^4}{64} dx - (3)^2$$

M1

$$= \left. \frac{3x^5}{320} \right|_0^4 - 9$$

on their $f(x)$ and μ^2

A1ft

$$= 9\frac{3}{5} - 9$$

$$= \frac{3}{5} \text{ or } 0.6$$

CAO

A1

3

[10]

Q26.

- (a) Use of area for probability
may be implied in (a) or (b)

M1

(i) $P(T < 10) = 0.9$
CAO

A1

(ii) $P(T > 5) = 0.6$
CAO

A1

3

(b) $P(T < 10 | T > 5) = \frac{P(5 < T < 10)}{P(T > 5)}$
attempt at conditional probability

M1

correct form

A1

$$= \frac{(i) - (1 - (ii))}{(ii)}$$

m1

$$= \frac{0.9 - (1 - 0.6)}{0.6} = \frac{0.5}{0.6}$$

$$= \frac{5}{6} \text{ Or } 0.833$$

CAO/AWRT

OR

(Conditional) area above 5 is $6a$
attempt at area above 5

(M1)

(Conditional) area above 5 must = 1
area must = 1 used

(m1)

Thus $a = \frac{1}{6}$

(A1)

Thus area below 10 is $(10 - 5)$

$= \frac{5}{6}$ Or 0.833

CAO/AWRT

(A1)

Note:

many candidates will quote answers
 without any working so:

3 correct 7 marks in total

2 correct in (a) 3 marks in (a)

1 correct in (a) 2 marks in (a)

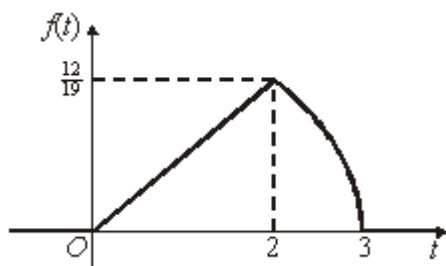
$\frac{x}{0.6} < 1$ in (b) 2 marks in (b)

$\frac{0.5}{0.6}$ in (b) 3 marks in (b)

[7]

Q27.

(a)



B1 for straight line on $[0, 2]$

B1 for curve on $[2, 3]$

B2

2

$$(b) \quad P(T < 0.5) = \frac{1}{2} \times \frac{1}{2} \times \frac{3}{19} = \frac{3}{76}$$

or by integration:

$$\int_0^{0.5} \frac{6t}{19} dt = \left[\frac{3t^2}{19} \right]_0^{0.5}$$

M1

$$= 0.0395$$

$$= \frac{3}{76} = 0.0395 \text{ (AG)}$$

A1

2

$$(c) \quad E(T) = \frac{6}{19} \int_0^2 t^2 dt + \frac{6}{19} \int_2^3 t^2 (3-t) dt$$

M1

$$= \left[\frac{2t^3}{19} \right]_0^2 + \left[\frac{6t^3}{19} - \frac{3t^4}{38} \right]_2^3$$

A1A1

$$= \frac{16}{19} + \frac{33}{38}$$

M1

$$= \frac{65}{38}$$

$$= 1.71$$

CAO

A1

5

[9]

Q28.

$$(a) \quad P(T < 8) = (8 - 4) \times \left(\frac{0.05 \times 0.1}{2} \right)$$

Trapezium

Worthwhile attempt at correct area, however divided

M1

or

$$\{(8-4) \times 0.05\} + \left\{ \frac{1}{2} \times (8-4) \times (0.1-0.05) \right\}$$

Rectangle + triangle

$$= 0.3$$

CAO; OE

A1

2

- (b) (i) Area under graph = 1

Use of; may be implied by their area;

accept $P(T > 8) = 1 - (a)$ must be stated clearly in reverse method

M1

$$\text{Area} = (a) + \{(s-8) \times 0.1\} + \left\{ \frac{1}{2} \times (20-s) \times 0.1 \right\}$$

Worthwhile attempt at area under given graph or area above 8, however divided

M1

or

$$\left(\frac{(20-8) + (s-8)}{2} \right) \times 0.1$$

Hence $0.05s = 0.5$
(implies $s = 10$)

CAO; OE

AG

NB: In reverse method, assuming

$s = 10$ so triangle area = 0.5 then showing

$s = 10$ given rectangle area = 0.2, scores

max of M1M1A0

A1

3

(ii) $P(T > 15) = \frac{1}{2} \times (20-15) \times f(15)$

$$\int_{15}^{20} y dx$$

Area of correct triangle or

M1

However using (b)

$$f(15) = 0.05$$



B1

$$\text{Thus } P(T > 15) = 0.125$$

CAO; OE

A1

3

[8]

Q29.

$$(a) \quad f(6) = \frac{2}{5} = 0.4$$

$$P(T \geq 6) = \frac{1}{2} \times \frac{3}{2} \times \frac{2}{5}$$

$$= \frac{3}{10} \text{ or } 0.3$$

M1

A1

2

$$(b) \quad \int_3^m \frac{1}{90} t^2 dt = 0.5$$

Correct limits

M1

$$\int f(t) dt = 0.5$$

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M1

$$\therefore \left(\frac{1}{270} t^3 \right)_3^m = 0.5$$

$$\int \frac{1}{90} t^2 dt$$

m1

$$= \frac{1}{270} t^3$$

A1

$$\left. \begin{array}{l} m^3 - 27 = 135 \\ m^3 = 162 \end{array} \right\}$$

m1

$$m = 5.45$$

CAO

A1f.t.

6

(c)
$$\int_3^6 \frac{1}{90} t^3 dt + \int_6^{7.5} \left(2t - \frac{4}{15} t^2 \right) dt$$

Attempt at:

M1

(i) $f(t)$

m1

(ii) two integrals

$$\left[\frac{t^4}{360} \right]_3^6 + \left[t^2 - \frac{4}{45} t^3 \right]_6^{7.5}$$

(iii) correct integration

A1A1

$$\begin{aligned} & (3.6 - 0.225) + (18.75 - 16.8) \\ & 3.375 - 1.95 \\ & = 5.325 \end{aligned}$$

$$\frac{13}{540} \text{ (AWRT 5.33)}$$

A1

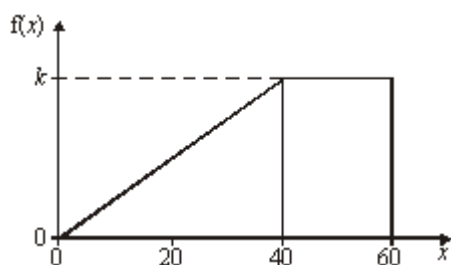
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5

[13]

Q30.

(a)



x -axis; (0) to 60

B1

$f(x)$ -axis; (0) to k or 0.025

B1

+ve slope straight line; 0 to 40

B1

horizontal straight line; 40 to 60
(allow **minor** extensions)
(0 for axes reversed)

B1

4

- (b) Area under graph = 1

use of; may be implied by their area **must be**
stated for $k = 0.025$ assumed

M1

$$\text{Area} = \left(\frac{1}{2} \times 40 \times k \right) + (20 \times k)$$

area of (triangle + rectangle)
[= 0.5 + 0.5 (+A1)]

M1

or

$$\text{Area} = k \times \left(\frac{60 + 20}{2} \right)$$

area of (trapezium)
[= 1 (+ A1)]

$$= 40k$$

cao; or equivalent

A1

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3

(implies $k = 0.025$)

AG

(Area = $40k \Rightarrow$ M0 M1 A1)

- (c) At $x = 30$ height = $0.75k$ or 0.0188

cao/awrt or equivalent

B1

$$\text{or } \left[\frac{kx^2}{80} \right]_0^{30}$$

or equivalent

$$P(X > 30) = \left(10 \times \left(\frac{0.75k + k}{2} \right) \right) + (20 \times k)$$

area of (trapezium + rectangle)

M1

$$\begin{aligned} & \text{or } 1 - \left(\frac{1}{2} \times 30 \times 0.75k \right) \\ & \quad 1 - \text{area of (triangle)} \\ & = 28.75k \quad \text{or } (1 - 11.25k) \\ & \quad 115k/4 \quad \text{or } (1 - 45k/4) \\ & = 23/32 \quad \text{or } 0.719 \\ & \quad \text{cao/awrt } (0.71875) \end{aligned}$$

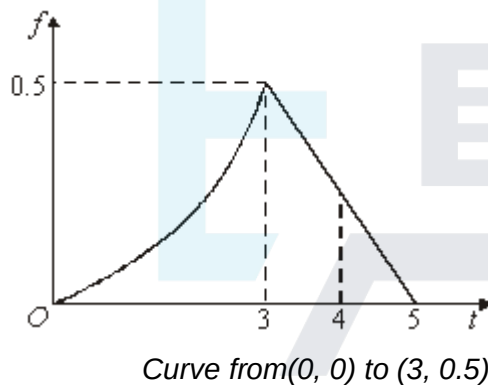
A1

3

[10]

Q31.

(a)



B1

Straight line from (3, 0.5) to (5, 0)

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B1

2

(b)

$$F(t) = \begin{cases} \frac{t^3}{54} & 0 \leq t \leq 3 \\ \frac{1}{8} (10t - t^2 - 17) & 3 \leq t \leq 5 \end{cases}$$

B1

M1M1

A1

4

(c) $P(T < 4) = F(4) = \frac{1}{8} (40 - 16 - 17)$

M1

$$\frac{7}{8} \text{ or } 0.875$$

Alternative (c): $1 - \frac{1}{2} \times 1 \times f(4)$

$$1 - \frac{1}{2} \times 1 \times \frac{1}{4}$$

$$1 - \frac{1}{8} = \frac{7}{8}$$

A1

2

[8]

Q32.

(a) Area = 1

Use of

$$\text{Area} = \frac{4 \times 2c}{2} + (2 \times 2c) + \frac{12 \times 2c}{2}$$

Attempt at area of (2 triangles + rectangle)
or equivalent

M1

M1

$$= 20c$$

CAO

A1

3

$$\therefore 20c = 1 \Rightarrow c = 0.05$$

AG

(b) (i) $P(X > 4) = P(4 < X < 6) + P(X > 6)$

M1

or $= 1 - P(X < 4)$

Attempt at area of (rectangle + triangle)
or (1 - triangle) or equivalent

$$= 4c + 12c = 1 - 4c = 0.8$$

CAO; or equivalent

A1

2

(ii) $P(4 < X < 12) = P(X < 12) - P(X < 4)$

use of; or equivalent

M1

$$= \{(1 - 3c) \text{ or } (4c + 4c + 9c)\} - (4c)$$

$$= (1 - 7c) \text{ or } 13c$$

either

A1

$$= 1 - 0.35 = 0.65$$

CAO; or equivalent

A1

3

$$(iii) \quad P(X < 12 | X > 4) = \frac{P(4 < X < 12)}{P(X > 4)}$$

Attempt at conditional probability

Correct expression

M1

A1

$$= \frac{(ii)}{(i)} = \frac{0.65}{0.80}$$

m1

$$= \frac{13}{16} = 0.8125$$

CAO or AWWF 0.812 to 0.813

A1

4

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NB Area > 4 is 16c so for conditional

(M1)

distribution, c = 0.0625

(A1)

Area < 12 is 13c for this distribution

(m1)

Thus probability = 13 × 0.0625 = 0.8125

(A1)

- (c) Some delays greater than 18 minutes
Some appointments early
PDF unlikely to be linear

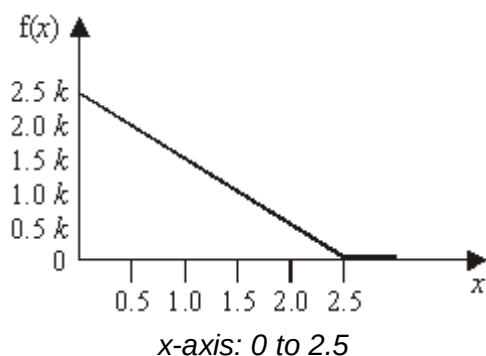
E1

1

[13]

Q33.

(a)



B1

*f(x)-axis: 0 to 2.5k (or 0.8);
assume starts at 0 if not stated*

B1

*straight line with negative slope;
ignore minor extensions;
(0, 2.5k or 0.8) to (2.5, 0)*

B1

3

(b) Area under line = 1

use of (may be implied)

M1

$$\text{Area} = \frac{1}{2} \times 2.5 \times 2.5k$$

Attempt at area (0 to 2.5)

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M1

$$\text{Or} = \int_0^{2.5} k(2.5 - x) dx$$

attempt at integration (0 to 2.5)

$$= 3.125k$$

(= 1 implies that k = 0.32)

CAO or equivalent

AG

NB accept reverse argument

A1

3

(c) $P(1 < X < 2)$:

$$f(1) = 1.5k = 0.48$$

$$f(2) = 0.5k = 0.16$$



CAO both may be implied

B1

$$\text{Area} = (2 - 1) \times \frac{(0.48 + 0.16)}{2}$$

Attempt at area of trapezium (1 to 2) or equivalent

M0 for $0.48 - 0.16 = 0.32$

M1

$$0.32 = 8/25$$

CAO

A1

or

$$\text{Area} = \int_1^2 0.32 (2.5 - x) dx$$

Attempt at area (1 to 2)

(M1)

$$\left[0.8x - 0.16x^2 \right]_1^2 =$$

Fully correct expression; allow switch of limits

(A1)

$$(1.6 - 0.64) - (0.8 - 0.16) = 0.32 = 8/25$$

CAO

(A1)

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3

[9]

Q34.

(a) $P(T = 4) = 0$

B1

1

(b) median = 3 $\Rightarrow P(T \leq 3) = 0.5$

$$P(T > 3) = \frac{1}{2} \times f(3) \times (5 - 3)$$

M1

$$\begin{aligned} &= \frac{1}{2} \times 0.5 \times 2 \\ &= 0.5 \end{aligned}$$

A1

2

$$(c) \quad P(T \geq 3.5) = \frac{1}{2} \times f(3.5) \times (5 - 3.5)$$

M1

$$= \frac{1}{2} \times \frac{3}{8} \times 1.5$$

M1A1

$$= \frac{9}{32}$$

$$0.28125$$

A1

$$\therefore P(T < 3.5) = 1 - \frac{9}{32}$$

$$= \frac{23}{32}$$

$$0.71875 \text{ (AWRT 0.719)}$$

A1

5

$$(d) \quad E(T) = \int_0^3 \frac{t^3}{18} dt + \frac{1}{4} \int_3^5 t(5-t) dt$$

M1

$$= \left[\frac{t^4}{72} \right]_0^3 + \left[\frac{1}{4} \left\{ \frac{5t^2}{2} - \frac{t^3}{3} \right\} \right]_3^5$$

Integration attempted

M1A1

$$= \frac{9}{8} + \frac{1}{4} \left\{ \left(\frac{125}{2} - \frac{125}{3} \right) - \left(\frac{45}{2} - 9 \right) \right\}$$

Substituting limits

M1

$$= 1 \frac{1}{8} + 1 \frac{5}{6}$$

$$= 2 \frac{23}{24}$$

A1

5

[13]

Q35.

- (a) Area under graph = 1

use of; may be implied by their area

Must be clearly stated in reverse argument

M1

$$\text{Area} = 20c + 2 \times 20 \times \left(\frac{c + 3c}{2} \right) = 20c + 80c$$

$$\text{or } = 60c + 2 \times \left(\frac{1}{2} \times 20 \times 2c \right) = 60c + 40c$$

worthwhile attempt at area under given graph, however divided

statement of form $100c = 1 \Rightarrow$ M1M0A0

Use of heights $\Rightarrow$ M0M0A0

M1

$$= 100c$$

$$(= 1 \text{ implies } c = 0.01)$$

CAO; stated or clear

AG

NB accept reverse argument

A1

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3

(b) (i) $P(X > 20) = 10 \times \left(\frac{2c + 3c}{2} \right) = 25c$

$$\text{or } = 10c + \left(\frac{1}{2} \times 10 \times (3c - 2c) \right) = 10c + 5c$$

identification of $2c$ or 0.02 and worthwhile attempt at correct area, however divided

M1

$$= 25c = \frac{1}{4} \text{ or } 0.25$$

CAO; OE

A1

2

$$(ii) \quad P(X < 10) = 20c + 20 \times \left(\frac{c + 3c}{2} \right) = 20c + 40c$$

$$\text{or } = 40c + \left(\frac{1}{2} \times 20 \times 2c \right) = 40c + 20c$$

M1

worthwhile attempt at correct area, however divided

$$\text{or } 1 - 20 \times \left(\frac{c + 3c}{2} \right) = 1 - 40c$$

$$\text{or } = 1 - 20c - \left(\frac{1}{2} \times 20 \times 2c \right) = 1 - 20c - 20c$$

$$= 60c = \frac{3}{5} \text{ or } 0.6$$

CAO; OE

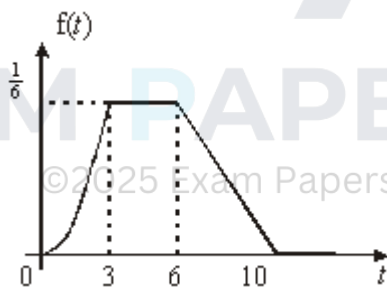
A1

2

[7]

Q36.

(a)



1 for each correct part of graph

B4

4

1 for axes $\left(\frac{1}{6} \text{ and } 10; f(t); t \right)$

(b) (i) $P(3 \leq T < 6) = \text{Area of rectangle}$

M1

$$= 3 \times \frac{1}{6}$$

$$= \frac{1}{2}$$

A1

2

(ii) $P(T > 6) = \text{Area of triangle}$

M1

$$= \frac{1}{2} \times \frac{1}{6} \times 4$$

$$= \frac{1}{3}$$

(accept 0.33)

A1

2

(iii) $\therefore P(T < 3) = 1 - \left(\frac{1}{3} + \frac{1}{2} \right) = \frac{1}{6}$

(seen anywhere) or $\int_0^3 f(t) dt = \frac{1}{6}$

B1

$$P(T < \text{lower quartile}) = \frac{1}{4}$$

$$P(T > Q_1) = \frac{3}{4} \text{ or } \int_0^{Q_1} f(t) dt = \frac{1}{4}$$

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$\therefore$ lower quartile lies in the range $3 < T < 6$

$\therefore$ lower quartile $= 3 + d$

$$\text{Area} = d \times \frac{1}{6} = \frac{1}{4} \times \frac{1}{6} = \frac{1}{12}$$

Any valid attempt at d

M1

$$\text{where } d = \frac{1}{12} \div \frac{1}{6} = \frac{1}{2}$$

$$\therefore \text{lower quartile} = 3 + \frac{1}{2} = 3.5$$

CAO

A1

4

$$(c) \quad P(T < 2) = \int_0^2 \left(\frac{1}{54} t^2 \right) dt$$

(ignore limits)

M1

$$= \left\{ \frac{1}{162} t^3 \right\}_0^2$$

(for $\frac{1}{162} t^3$; OE)

A1

$$= \frac{4}{81}$$

$$= 0.0494 \text{ (3sf)}$$

$$= 5\%$$

$$= \frac{4}{81}$$

$$= 0.0494 \text{ (3sf)}$$

$$\approx 5\%$$

(Substitution of correct limits)

m1

any correct form (AWRT 0.049)

A1

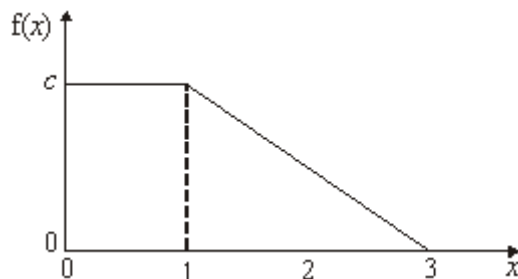
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[16]

Q37.

(a)



x -axis: 0 to 3

B1

$f(x)$ -axis; 0 to c

B1

horizontal line: 0 to 1 at c

B1

negative slope line;

B1

4

(1, c f.t.) to (3, 0)

- (b) Area under graph = 1
Use of
May be implied by their area

M1

$$\text{Area} = c + \frac{2 \times c}{2} \text{ or } c \times \left(\frac{3+1}{2} \right)$$

Area of (rectangle + triangle)
or
Area of (trapezium)

M1

$$= 2c$$

$$\text{Thus } 2c = 1 \text{ implies } c = \frac{1}{2} \text{ or } .0.5$$

CAO

A1

3

- (c) $f(2) = c/2 = \frac{1}{4}$ or 0.25
ft on c; may be implied

B1f.t.

$$P(X < 2) = 1 - P(X > 2)$$

$$= 1 - \frac{1 \times (c/2)}{2} = 1 - \frac{c}{4} = 1 - \frac{1}{8}$$

1 – area of triangle

Or

M1

$$= c + 1 \times \left(\frac{c + (c/2)}{2} \right) = \frac{7c}{4} = \frac{7}{8}$$

Area of (rectangle + trapezium)

$$= \frac{7}{8} \text{ or } 0.875$$

CAO

A1

3

[10]

Q38.

(a) $f(x) = \frac{1}{6}, 6 \leq x \leq 12$

Mean, $\mu = \frac{6+12}{2} = 9$

CAO

B1

Variance, $\sigma^2 = \frac{(12-6)^2}{12} = 3$

CAO

B1

2

(b) $P(X > \mu + \sigma) = P(X > 9 + \sqrt{3})$

Concept of area of a rectangle or correct integral

M1

$$= \frac{12 - (9 + \sqrt{3})}{6}$$

or $1 - \frac{(9 + \sqrt{3}) - 6}{6}$

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f.t. on (a): must use $\sqrt{\quad}$ or equivalent
Wrong tail $\Rightarrow$ M1A1 f.t. A0 max
or correct integration

A1f.t.

$= 0.21$

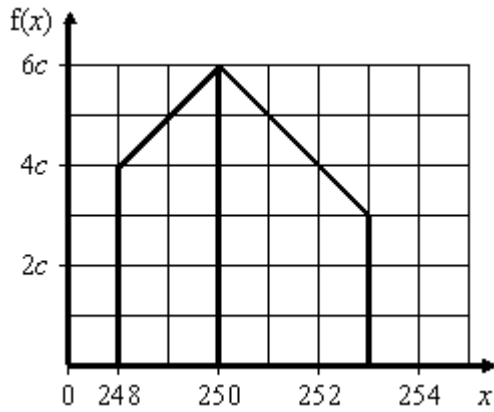
AWRT; accept $\frac{3 - \sqrt{3}}{6}$ OE

A1

3

[5]

Q39.



In parts (a) and (c), accept answers based on counting squares and triangles

(a) Area = 1

Use of

M1

$$\text{Area} = \frac{2(4c + 6c)}{2} + \frac{3(6c + 3c)}{2}$$

Attempt at area

≥ 1 correct area even if split into 2 rectangles and 2 triangles

M1/A1

$$= 10c + \frac{27c}{2} = \frac{47c}{2}$$

Area = 23.5c or equivalent

A1

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4

$$\therefore c = \frac{2}{47}$$

AG

(b) Zero

CAO

B1

1

(c) $P(249 < X < 252) =$

Correct area(s) identified and attempted
Accept split into triangles and rectangles.

M1

Either:

$$P(249 < X < 250) + P(250 < X < 252) =$$

$$\frac{(5c + 6c)}{2} + \frac{2(6c + 4c)}{2} =$$

$$55c + 10c = 15.5c$$

or:

$$1 - \frac{(4c + 5c)}{2} - \frac{(4c + 3c)}{2} =$$

$$1 - 4.5c - 3.5c = 1 - 8c$$

Correct expression

A1

$$= \frac{31}{47} = 0.66$$

CAO/AWRT

A1

3

[8]

Q40.

(a) Area = 1 = $\frac{1}{2} \times 4 \times 4k$

$$\int_0^4 f(x) dx = 1$$

M1

$$8k = 1$$

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$$k = \frac{1}{8}$$

AG

A1

2

(b) $E(X^2) = \int_0^4 x^2 \cdot \frac{1}{8} x dx$

M1

$$= \left[\frac{1}{32} x^4 \right]_0^4$$

A1

$$= 8$$

A1

$$\text{Var}(X) = E(X^2) - E(X)^2$$

$$= 8 - \left(2\frac{2}{3}\right)^2$$

$$= \frac{8}{9}$$

ft on their E(X²)
OE

A1f.t.

4

(c) $\text{Var}(5X - 4) = 25 \text{Var}(X)$

$$= 25 \times \frac{8}{9}$$

for 5² × their Var(x)
5² × Var(x) ± a
⇒ M0A0

$$= 22\frac{2}{9}$$

OE

A1f.t.

2

(d) Let $a = 90^{\text{th}}$ percentile

$$\text{then } \int_0^a \frac{1}{8} x \, dx = 0.9$$

$$\left[\frac{1}{16} x^2 \right]_0^a = 0.9$$

$$\frac{1}{16} \times a^2 = 0.9$$

any of these seen or
 $\int_a^4 \frac{1}{8} x \, dx = 0.1$

M1

$$a^2 = 0.9 \times 16 = 14.4$$

for 14.4 seen

A1

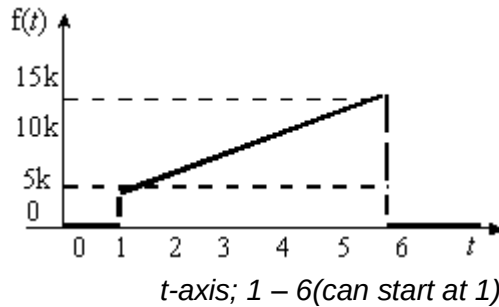
$$a = 3.79(3\text{sf})$$

$$3.7947$$

$$(\pm 3.79 \Rightarrow \text{M1A1A0})$$

Q41.

(a)



$f(t)$ axis; 0 to 15k or 0 to 0.3

(assume 0 if not stated)

straight line;

(1, 5k) to (6, 15k) or (1, 0.1) to (6, 0.3);
ignore minor extensions

(b) Area = 1

use of (may be implied)

Area = $5 \times \left(\frac{5k + 15k}{2} \right)$

attempt at area of trapezium or rectangle + triangle

or $\int_1^6 k(2t + 3) dt$

= $(5 \times 5k) + \left(\frac{1 \times 5 \times 10k}{2} \right)$

= 50k

CAO; AG

(= 1 $\Rightarrow k = 1/50$)

- (c) (i) $f(3) = 9k$ or 0.18
CAO

B1

$$P(T < 3) = (3 - 1) \left(\frac{5k + 9k}{2} \right)$$

attempt at correct area, however divided or attempted integration (1,3) with A1 for $k(t^2 + 3t)$ and limits (1,3)

M1

$$= (2 \times 5k) + \left(\frac{1 \times 2 \times 4k}{2} \right)$$

$$= 14k = 0.28$$

CAO OE

A1

3

- (ii) $f(2) = 7k$ or 0.14

CAO both

B1

$$f(5) = 13k \text{ or } 0.26$$

$$P(2 < T < 5)$$

$$= (5 - 2) \left(\frac{7k + 13k}{2} \right)$$

attempt at correct area, however divided or attempted integration (2,5) with A1 for $k(t^2 + 3t)$ and limits (2,5)

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M1

$$30k = 0.6$$

CAO OE

$$\text{NB } 1 - 0.14 - 0.26 = 0.6 \Rightarrow \text{B1M0A0}$$

A1

3

[12]

Q42.

(a) $\int_1^5 ks^2 \, ds = 1$
for $\int ks^2 \, ds$

M1

$$\Rightarrow \left[\frac{ks^3}{3} \right]_1^5 = 1$$

for $\frac{ks^3}{3}$

A1

$$\Rightarrow \frac{124}{3}k = 1$$

$$k = \frac{3}{124}$$

AG

A1

4

$$(b) \quad P(S < 3) = \frac{3}{124} \int_1^3 s^2 ds$$

$$1 - \int_3^5 ks^2 ds$$

M1

$$= \left[\frac{s^3}{124} \right]_1^3$$

(limits must be correct for M1)

$$= \frac{27}{124} - \frac{1}{124}$$

$$= \frac{26}{124}$$

$$\text{substitute limits } 1 - \left(\frac{125}{3} - \frac{27}{3} \right)k$$

$$= 1 - \frac{49}{62} = \frac{13}{62}$$

m1

$$= \frac{13}{62}$$

AWRT 0.210

A1

3

(c) $\frac{3}{124} \int_1^{\infty} s^2 ds = 0.5$
 $\int_k^5 s^2 ds = 0.5$

M1

$\left[\frac{s^3}{124} \right]_1^{\infty} = 0.5$
 $\left[\frac{s^3}{124} \right]_k^5 = 0.5$

A1

$m^3 - 1 = 62$
 $125 - m^3 = 62$

M1

$m^3 = 63$
 $m = \sqrt[3]{63}$
 $m = 3.98$ (3SF)
 $m = 4$ min

A1

4

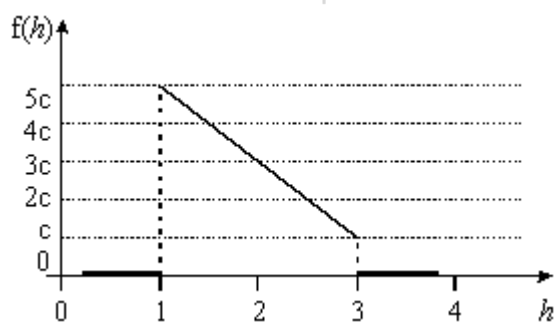
[11]

EXAM PAPERS PRACTICE

Q43.

(a)

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h -axis: 1 to 3; can start at 1

B1

$f(h)$ -axis: 0 to $5c$ or 0 to $\frac{5}{6}$;

B1

assume starts at 0 if not stated

B1

Straight line, negative slope which must not meet h axis;

ignore minor extensions (1,5c) or $\left(1, \frac{5}{6}\right)$ & (3,c) or $\left(3, \frac{5}{6}\right)$

3

(b) Area under line = 1

Use of (may be implied)

M1

$$\text{Area} = (3-1) \times \frac{5c+c}{2}$$

Attempted at area of trapezium

$$= [(3-1) \times c] + \left[\frac{1}{2} \times (3-1) \times (5c-c) \right]$$

Attempt at area of rectangle plus triangle

M1

$$= \int_1^3 c(7-2h)dh$$

Attempt at integration

$$= 6c$$

CAO

A1

$$\left(= 1 \text{ implies } c = \frac{1}{6} \right)$$

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3

(c) Area = $(3-2) \times \frac{(3c+c)}{2}$

$$= [(3-2) \times c] + \left[\frac{1}{2} \times (3-2) \times (3c-c) \right]$$

Attempt at correct area

M1

$$= \int_2^3 c(7-2h)dh$$

$$= 2c = \frac{1}{3}$$

CAO; accept AWRT 0.333

A1

2

(d) $P\left(2 < H < 2\frac{1}{2}\right) =$

Attempt at

M1

$$\frac{1}{2} \times \left(\frac{3c + 2c}{2}\right) = \frac{5c}{4} = \frac{5}{24} = 0.208$$

CAO/AWRT

A1

$$P\left(H < 2\frac{1}{2} \mid H > 2\right) = \frac{P\left(2 < H < 2\frac{1}{2}\right)}{P(H > 2)}$$

Attempt at conditional probability

M1

$$= \frac{\frac{5}{24}}{\text{part (c)}}$$

m1

$$\frac{5}{8} = 0.625$$

CAO

A1

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5

[13]

Q44.

(a) (i) 1

1 cao

B1

(ii) 1.54

1.54 cao

B1

(iii) 1.56

1.56 cao

B1

(iv) 1.73
1.73 cao

B1

(v) 1
1 cao

B1

5

(b) (i) $0.5 - 0.291 = 0.209$
method

M1

0.209 cao

A1

2

(ii) $\sqrt{0.0751} = 0.2474$
method

M1

0.274 (0.2735 to 0.2745)

A1

2

allow B1 for variance = 0.0751

(iii) $E(X^2) - 1.54^2 = 0.0751$
any correct equation

M1

$E(X^2) = 2.45$

2.45 (2.44 to 2.45)

A1

2

[11]

Q45.

(a) $\int_0^1 px + q \, dx$

Correct expression - ignore limits

M1

$\left[\frac{px^2}{2} + qx \right]_0^1 = 1$

Any correct integration - anywhere

M1

$$\frac{p}{2} + q = 1$$

Correct prrof - **ag** - allow by diagram

A1

3

(b) $E(X) = \int_0^1 px^2 + qx \, dx$

Correct expression - ignore limits

M1

$$= \left[\frac{px^3}{3} + \frac{qx^2}{2} \right]_0^1$$

Correct method

m1

$$= \frac{p}{3} + \frac{q}{2}$$

$$\frac{p}{3} + \frac{q}{2} \text{ cao}$$

A1

3

(c) $\frac{p}{3} + \frac{q}{2} = 0.6$

Correct equation – candidate's (b)
(assuming $p = 1.2$ and showing $q = 0.4$ M0)

M1

$$q = 1 - \frac{p}{2} \text{ from (a)}$$

$$\frac{p}{3} + 0.5 - \frac{p}{4} = 0.6$$

$$\frac{p}{12} = 0.1$$

Completely correct method

m1

from (a) $q = 0.4$

$p = 1.2$ **ag** $q = 0.4$ cao

A1

3

(d) $E(X^2) = \int 1.2x^3 + 0.4x^2 dx$

Correct expression for $E(X^2)$, candidate's p and q - allow in terms of p and q and if called variance

M1

$$= \left[\frac{1.2x^4}{4} + \frac{0.4x^3}{3} \right]_0^1$$

Correct expression for $E(X^2)$, candidate's p and q - allow in terms of p and q and if called variance

m1

$$= 0.4333$$

$$V(X) = 0.4333 - 0.6^2 = 0.07333$$

Completely correct method for σ - allow variance if called variance

m1

$$\sigma = 0.271$$

$$0.271(0.27 - 0.272)$$

A1

4

[13]

EXAM PAPERS PRACTICE

Q46.

(a) $\int_0^2 cx + d dx = 1$

Any correct expression - ignore limits

M1

$$\left[\frac{cx^2}{2} + dx \right]_0^2 = 1$$

Any correct integration - may be awarded elsewhere in question

M1

$$2c + 2d = 1$$

Wholly correct proof allow geometrical proof

A1

3

(b) (i) $\int_0^m 0.5 \, dx = 0.5$

Any correct expression

M1

$$\left[0.5x \right]_0^m = 0.5$$

$$m = 1$$

1 **cao** by correct method allow geometrical proof

A1

2

(ii) $\int_0^m 0.5x \, dx = 0.5$

$$m = 1$$

Any correct expression

M1

$$\left[\frac{0.5x^2}{2} \right]_0^m = 0.5$$

Any correct equation after integration

M1

$$m = \sqrt{2} = 1.41$$

$\sqrt{2}$ or 1.41 (1.41 to 1.42) disallow $\pm \sqrt{2}$

A1

3

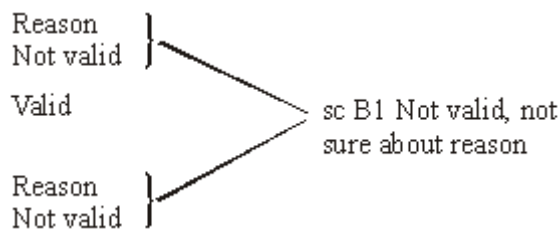
(c) **A** not valid - $2c + 2d \neq 1$

B valid

C not valid = $f(x)$ negative for small x

M1

5



A1

B1

M1

A1

[13]

Q47.

- (a) Maximum of p.d.f.

Maximum of graph

E1

1

- (b) (i) Variance = $0.369 - 0.542^2 = 0.075236$

Method for variance

M1

$$\text{s.d.} = \sqrt{0.075236} = 0.274$$

Method for s.d. - disallow if called variance

m1

*0.274 (0.274 – 0.275)
SC allow A1 for variance =
0.0752 (0.0752 – 0.0753)*

A1

3

- (ii) $1 - 0.719 = 0.281$

Use of 0.719

0.281 cao

B1B1

2

- (iii) 0.557

0.557 cao

B1

1

- (iv) $0.5 - 0.483 = 0.017$

Use of 0.483, allow $1 - 0.483$

B1

Correct method

M1

0.017 cao

A1

3

[10]

Q48.

- (a) $\int_{-6}^{14} k \, dy = [ky]_{-6}^{14} = k(14 - (-6)) = 20k$

Allow by diagram – generous

M1

$$20k = 1 \quad k = 0.05$$

0.05 ag

A1

2

(b) 4

4 cao

B1

1

(c) (i)

$$\int_{-6}^{14} 0.05y^2 dy = \left[\frac{0.05y^3}{3} \right]_{-6}^{14}$$

Any correct expression for $E(Y^2)$ – ignore limits
 Correct method – arithmetic/sign errors only

M1 m1

$$= 45.733 - (-3.6) = 49.3$$

$$49.333 (49.3 - 49.35)$$

A1

3

(ii) $\text{Var}(Y) = 49.3333 - 4^2 = 33.3333$

Method for Var – their $E(Y^2)$

M1

$$\text{s.d.} = \sqrt{33.3333} = 5.77$$

Requires all previous method marks in (c)
 $5.77(5.77 - 5.775)$

m1 A1

3

(d) (i) 0.3

0.3 acf

B1

1

(ii)

$$\int_{-2}^2 0.05 dy = [0.05y]_{-2}^2$$

Allow diagram

M1

$$= 0.2$$

0.2 ag

A1

2

(iii) 0.6

Allow diagram
0.6 acf

M1 A1

2

[14]



EXAM PAPERS PRACTICE

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Examiner reports

Q1.

The majority of students did not score the mark for this question. Most students simply substituted $x = 1$ into the provided probability density function to obtain 34. Some students either integrated the probability density function between 0 and 1 or between 1 and 2 to obtain $\frac{3}{4}$. Some students either integrated the probability density function between 0 and 1 or between 1 and 2 to obtain $\frac{1}{2}$.

Q2.

Part (a) was answered correctly by the vast majority of students. The most common errors were to either write 'continuous random variable' or to give an interval for the mode. Some students attempted the probability density function here and did not attempt part (b).

Part (b) was the most challenging on the paper with many students scoring zero marks or not responding at all. The most common response scoring zero marks was to write down the probability density function with either gaps left for the two straight lines or constants written there instead. A significant number of students understood that they needed to find the y coordinate of the maximum point of the probability density function but could not develop their solution any further. Many students managed to find the straight line for the first interval but made errors in attempting to find the second. The most common error for finding the straight line for the second interval was to assume that it passed through the origin. Some students who successfully found the two straight lines did not define the probability density function as zero for all other values.

Q5.

This was another well-answered question, with many students scoring full marks. A plot was not required for the sketch (though many did produce one) but an indication of scale on both axes was required. Students should appreciate the distinction between a requirement to 'Show' as in part (b), where each step of the working is expected, and 'Write down' as in part (c)(i) where only the answer is needed. Having shown that the area splits neatly in half as $9k + 9k$, many students wasted time with up to a page of working to find the median value.

Q6.

In attempting this question, the population of students split essentially into two parts. The major part understood the concept of $F(t)$ and its relationship to $f(t)$ and generally scored very highly. The minor part usually started by integrating $F(t)$ and thereafter gained very few marks. Amongst the better students, it was disappointing that so many failed to convert 5.79 weeks into days, as required, and equally that so many at this level differentiated t^3 and obtained $2t^2$.

Q7.

The sketch was reasonably well done and most students correctly showed the integration from 0 to x to find $F(x)$. The calculation of the lower quartile was also well done. In part (c)(i), there was much fudging of the answer after the integral from 1 to x gave -4 while the required answer was -3 , causing many to lose the mark for the correctly evaluated

integral when they spuriously altered it. Often an incorrect answer to part (i) was carried over into part (ii) giving an incorrect equation, with the given $F(x)$ been ignored. It was disconcerting to see how many students at this level could not even recognise or subsequently solve a quadratic equation. On the other hand, many students scored highly on all parts of this question.

Q8.

Students should not have expected to gain full marks when they failed to use a ruler to help them draw the straight lines required to complete the graph. *Freehand sketches* were not good enough since they were asked to *draw* a graph. Part (b) was usually completed successfully. Although students produced some excellent solutions to part (c), these were not always fully explained. Only the very best students managed a fully correct solution to all parts of (c). Many, in part (c)(iii), simply multiplied together their answers to parts (c)(i) and (c)(ii), obviously believing that the events $X > 2.5$ and $1.5 < X < 4.5$ were independent. This was incorrect and so gained no credit. Many students used calculus to find the required areas, often with more success than those who attempted to use a combination of areas of simple geometric shapes (rectangle, trapezium, triangle). Students using the latter method seemed unable, on the whole, to cope with calculations

involving fractions, often writing $\frac{5}{12}$ as 0.41 or 0.416 before attempting to calculate the required area. Marks were often lost, especially in parts (c)(iii) and (c)(iv), due to students' inability to write their answers in an **exact** form as requested. The correct answer to part

(c)(iii) of $\frac{17}{48}$ was often seen as the answer to part (c)(iv). Some students gave excellent solutions to this question by first finding the cumulative distribution function $F(x)$. Unfortunately, some weaker students who attempted this method were unable to find the correct form for $F(x)$ and so lost marks.

Q9.

The graph in part (a) was usually adequately sketched. There was, however, a minority of candidates who interpreted the request for a 'sketch' as an excuse to offer up a very poor quality freehand drawing. This should not have been the case and any such attempts were likely to be penalised.

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The majority of candidates used simple areas of rectangles to obtain their very efficient solutions to parts (b)(i) and (b)(ii). Most candidates realised that, when the answers are given in the question, sufficient working must be shown in order to obtain full credit. Some candidates, when they are confronted by a continuous distribution, seem to think that integration methods always have to be used. This is not the case when straight lines form part of the graph of the probability density function. Those candidates who used valid integration methods usually gained the correct given values and thus full credit.

Candidates were expected to use the information given in part (b) to write down the answers to parts (c)(i) and (c)(ii).

Although some candidates recognised that the answers to part (b) implied that the upper

quartile of X was $8\frac{1}{3}$ and that the lower quartile of X was 3, many attempted to recalculate these values, usually incorrectly, and mostly by the use of calculus methods.

Disappointingly, in part (d), only the most able candidates recognised that, whatever the values obtained for the median and lower quartile, $P(X < \text{median} \mid X \geq \text{lower quartile}) =$

$\frac{1}{3}$ and $P(X < \text{median and } |X \geq \text{lower quartile}) = \frac{1}{4}$ and are always true.

Q10.

Surprisingly, this was the worst attempted question on the paper. The majority of candidates gained both marks in part (a), but many failed to recognise that the continuous random variable, T , followed a rectangular distribution. Consequently, there were many attempts using calculus to find the values and, although these were often correct, this wasted time that could perhaps have been better used in answering other more difficult questions.

Part (b) was only answered correctly by the better candidates. Many did not understand that $P(-2 < T < 2)$ was required and so simply attempted to evaluate $P(T > 2)$ with the result that $23/30$ was the most prevalent incorrect answer. Too many candidates treated T as a discrete random variable with $P(T > 2) = 1 - P(T \leq 1)$ often seen, whilst others again used unnecessary integration rather than simple areas of rectangles.

Q11.

In the previous Report on the Examination, it stated that too many candidates seemed to be under the misapprehension that a request for a 'sketch' entitled them to draw a very poor quality graph or copy some unscaled diagram taken from the display on their graphics calculators. This still seemed to be the case and all such attempts lost some or all of the marks available. The axes should have been drawn with the aid of a ruler and pencil and, as a minimum, should have contained the appropriate scales; $O(0, 0)$, $(1, 0)$ and $(2, 0)$ on the x -axis and $(0, 0.5)$ and $(0, 1)$ on the y -axis. The two concave arcs were often seen as being convex or as straight lines.

In part (b), there were many correct solutions with $\int_0^1 \frac{1}{2}(x^2 + 1)dx = \frac{2}{3}$ often seen. Unfortunately, those candidates who drew straight lines in their sketches, usually ended up with $P(X \leq 1) = \frac{1}{2}(0.5 + 1) = \frac{3}{4}$; this gained no credit.

Part (c) was answered very well by many candidates with the standard of algebra and integration showing a marked improvement. However, it should be noted that, where an answer is given on the examination paper, sufficient method must be shown to ensure that full marks are awarded.

Part (d)(i) was answered very well by many candidates. However, some candidates,

having found that $\text{Var}(X) = \frac{4}{5} - \left(\frac{19}{24}\right)^2 = \frac{499}{2880}$, then were unable to write down the value of k as 2880. Whilst most candidates obtained the correct answer in part (d)(ii), a minority

thought incorrectly that $E(X^2) = \left(\frac{19}{24}\right)^2$ even though the correct value of $\frac{4}{5}$ was given in the question. Part(d)(iii) was quite well attempted although some candidates used $\text{Var}(12X - 5) = 12 \times \text{Var}(X)$ or even $\text{Var}(12X - 5) = 12^2 \times \text{Var}(X) - 5$.

Q12.

Part (a) was answered very poorly by too many candidates. Candidates were asked to **state** values for the median and lower quartile of X . This should have indicated to candidates that the answers required little work, but this was often not picked up. Some candidates obtained the correct values but these were often invalidated by incorrect methods. Those candidates who employed calculus often had more success, as they realised that $F(m) = 0.5$ and $F(q_1) = 0.25$ were required.

Part (b) was often done well by those candidates who realised that $F(x) = 1/54(x - 4)^3 + c$, as they usually went on to use either $F(1) = 1/2$ or $F(4) = 1$ to show that $c = 1$. Some candidates, having been given the answer in the question, simply wrote this down at the end of what was usually a totally incorrect method.

Part (c) was usually done by correctly attempting to evaluate $F(3) - F(2)$ but the arithmetic then defeated many.

Most candidates realised that $F(q) = 3/4$ was required and consequently went on to gain full marks for part (d)(i). The correct answer to part (d)(ii) was often seen, even by those candidates who could not do part (d)(i). However, some failed to give the answer to three decimal places and so lost the mark.

Q13.

A number of candidates seemed to have the wrong perception that the request for a 'sketch' entitled them to draw a very poor quality graph or copy some unscaled diagram which they had managed to take from their graphics calculator display. All such attempts lost some or all of the marks available. The axes should ideally have been drawn with the aid of a ruler and pencil and should have contained the appropriate scales; $O(0, 0)$, $(2, 0)$ and $(5, 0)$ on the x -axis and $(0, 0.5)$ on the y -axis as a bare minimum. A concave curve from $O(0, 0)$ to $(2, 0.5)$ and a straight line from $(2, 0.5)$ to $(5, 0)$ should then have been drawn.

Most candidates realised that $\int_2^x \frac{1}{6}(5 - x)dx$ was required somewhere in part (b) even though some were unsure of the limits and what this integral gave them. Some thought incorrectly that $F(x) = \int_2^x \frac{1}{6}(5 - x)dx \Rightarrow F(x) = -\frac{1}{12}(5 - x)^2 + c$, followed by $F(5) = 1 \Rightarrow c = 1$, and hence that $F(x) = 1 - \frac{1}{12}(5 - x)^2$ as required. This was very plausible but,

unfortunately, the original assumption that $F(x) = \int_2^x \frac{1}{6}(5 - x)dx$ was false! However, that said, there were some excellent and fully correct solutions to part (b) where the correct expression $F(x) = F(2) + \int_2^x \frac{1}{6}(5 - x)dx$ or $F(x) = \int_2^x \frac{1}{6}(5 - x)dx + k$ was used.

In part (c), the great majority of candidates were able to show that $F(4) = \frac{11}{12}$ and $F(3) = \frac{2}{3}$, and then went on to show that $P(3 \leq X \leq 4) = \frac{1}{4}$. However, some candidates treated this as if it were a discrete distribution, stating that $P(X \geq 3) = 1 - P(X \leq 2)$. Unfortunately, all but the very best candidates assumed incorrectly that $P(X \geq 3 | X \leq 4) = P(3 \leq X \leq 4) =$

$$\frac{1}{4}. \text{ Conditional probability was required using the fact that } P(X \geq 3 | X \leq 4) = 1 - \frac{F(3)}{F(4)} = 1 - \frac{8}{11} = \frac{3}{11} \text{ or } P(X \geq 3 | X \leq 4) = \frac{F(4) - F(3)}{F(4)} = \frac{\frac{1}{4}}{\frac{11}{12}} = \frac{3}{11}.$$

Q14.

In part(a), too many candidates seemed to have the wrong perception that the request for a 'sketch' entitled them to draw a very poor quality graph or copy some unscaled diagram which they had managed to take from the display on their graphics calculators. All such attempts lost some or all of the marks available. The axes and straight lines should have been drawn with the aid of a ruler and should have contained the appropriate scales: O or $(0, 0)$, $(1, 0)$ and $(3, 0)$ on the x -axis and $(0, 0.5)$ on the y -axis as a bare minimum. A straight line from $(0, 0.5)$ to $(1, 0.5)$ and a straight line from $(1, 0.5)$ to $(3, 0)$ should then have been drawn. It was often quite difficult to decide whether the sketch contained the required 'straight' lines or curves, so badly attempted were the sketches.

Similarly, part (b) was very poorly attempted with most candidates simply showing that $f(1) = 0.5$ or solving $f(x) = 0.5$ to give $x = 1$. Both of these were inappropriate and so gained no credit.

There were many good attempts at finding $E(X)$ in part (c). However when, as here, the answer is given in the question, sufficient relevant working must be shown for full credit to be obtained.

Although there were many excellent solutions to part (d), there were only a few candidates

who used the most efficient method of $1 - \frac{1}{2} \times 0.75 \times \frac{3}{16}$; most chose an alternative method which required integration. There were a large number of usually less able

candidates who wrote $P\left(X > 2\frac{1}{4}\right) = f\left(2\frac{1}{4}\right) = \frac{3}{16}$, thus gaining no marks.

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Q15.

In general, the attempts at this question were very poor. It should be emphasised, yet again, that when an answer is given in a question, candidates must take extra care to explain what they are doing. This was often not seen in answers to parts (a)(ii), (a)(iii) and (b)(i), where many candidates showed insufficient working to achieve full credit.

Part (a)(i) asked candidates to sketch the graph of F . Many interpreted this as allowing the drawing of a rather scruffy freehand diagram without a scale. This was not what was intended. It was expected that scaled (approximately) and labelled axes be drawn (using a ruler and preferably a pencil) and care be taken in drawing the curved part of the sketch.

Although most candidates attempted the straight line from the origin to $(1, \frac{1}{2})$ correctly, there was much less success with the curve from $(1, \frac{1}{2})$ to $(4, 1)$. The part of the graph defined by $F(x) = 1$ was often not drawn, and the part defined by $F(x) = 0$ was rarely seen at all.

Part (a)(ii) was not explained well by many candidates. Many thought incorrectly that dividing 4 (the length of the interval over which most of them drew their sketch) by 4 would give them the value of the lower quartile, whilst others assumed that (the value of the

lower quartile) = $\frac{1}{2} \times$ the value of the median) without any explanation as to why this was the case for this distribution.

It was, however, pleasing to see good candidates state correctly that $F(q_1) = \frac{1}{2}q_1 = 0.25$,

leading to the given answer of $q_1 = \frac{1}{2}$. A variety of methods, mostly correct, were employed in part (a)(iii). The vast majority of candidates realised that the upper quartile was in the interval $[1, 4]$ and used the fact that $F(q_3) = 0.75$ to good effect. Most used different rearrangements of the resultant cubic equation, whilst others used their calculators to show that $q_3 = 1.62$ (3 sf), to achieve the required given result.

The attempts at part (b)(i) were varied and mostly disappointing, with many candidates

using the result $\alpha = \frac{1}{2}$ to show that $\beta = \frac{1}{18}$ without firstly explaining why they thought that

$\alpha = \frac{1}{2}$. Some found only one relationship between α and β (usually either $\alpha = 9\beta$ or $\alpha + 9\beta = 1$) and then simply substituted the given values of α and β to show that these values did work. This was not felt to be sufficient to gain full credit since each of these equations, taken in isolation, had infinitely many solutions, of which the given solution was just one such pair of values. However, there were a few excellent solutions, the most successful of

these first showing that on the interval $[0, 1]$, $f(x) = \alpha = F'(x) = \frac{1}{2}$. This was then followed by use of the property that f was continuous at $x = 1$ to give $f(1) = \alpha = 9\beta$, thus giving $\beta = \frac{1}{18}$ as required.

Part (b)(ii) was a good source of marks, even for those candidates who gained virtually no credit on earlier parts of the question. These candidates showed good algebraic and integration skills in finding correctly that $E(X) = 1.125$. A small minority of candidates, instead of using any integration methods, used their calculators to find the correct

numerical answer, usually after stating correctly that $E(X) = \int_0^1 \frac{1}{2}x dx + \int_1^4 \frac{1}{18}x(x-4)^2 dx$.

This method gained the full 5 marks for those candidates who stated the correct numerical value but only 1 mark for those who did not. Whilst the use of calculators is encouraged in data analysis and in the evaluation of numerical expressions, candidates run the risk of losing marks if they acquire an incorrect numerical answer without any intermediate working.

Q16.

In part (a), many candidates were able to sketch the graph of f correctly. However, 'sketch the graph' does not give candidates licence to produce rough, freehand diagrams devoid of scales. In many cases, it was very difficult to give candidates credit since it was impossible to infer whether they intended to draw straight lines or curves. Consequently, some candidates lost some or all the marks for this part of the question. In part (b)(i),

many candidates used $\int_0^2 \frac{2}{15}t dt$, usually gaining the correct answer of $\frac{4}{15}$. However, as the graph contained straight lines, integration, although correct, was not the most efficient method. Although there were many correct solutions seen to part (b)(ii) by a variety of

correct methods, there were still some candidates who treated $f(t)$ as a discrete distribution and consequently thought incorrectly that $P(2 < T < 4) = P(T \leq 3) - P(T \leq 2)$. This method gained no credit. Some candidates, presumably from experience on past papers, first derived the cumulative distribution function, $F(t)$. Although this method gained the better candidates full marks, it was not the most efficient way of doing this part of the question. In part (c), the great majority of candidates considered the correct integrals and performed correct integration and were then able to evaluate these using correct limits to

gain the required answer of $2\frac{2}{3}$. However, some candidates were unable to substitute the limits correctly in order to obtain this correct answer. Unfortunately, those candidates who thought $f(t)$ to be discrete earlier in the question continued with this incorrect premise here.

Q17.

This was the least-well-answered question on the paper with many major errors of technique and understanding. It was only answered well by the more able candidates. It was a major concern that so many candidates thought incorrectly that

$E\left(\frac{1}{X}\right) = \frac{1}{E(X)}$ and $\text{Var}\left(\frac{1}{X}\right) = \frac{1}{\text{Var}(X)}$. These candidates wasted a lot of time evaluating $E(X)$ and $E(X^2)$ to no avail. There were also many examples of poor integration

techniques. When $E\left(\frac{1}{X}\right) = \int_0^1 \frac{1}{x} \times 3x^2 dx$ was correctly stated, some candidates then

$$\frac{3x^2}{3} \text{ or } \frac{3x^3/3}{x^2/2}$$

seemed unable to integrate correctly, with $\ln x \times$ often seen.

The fact that the variance cannot be less than zero seemed to have escaped the attention of some candidates who were apparently quite comfortable, not only in obtaining negative values for their variance, but also in using such values in part (b), where it was also evident that many candidates were let down by poor algebraic skills. It was expected that

candidates would first express $\frac{5+2X}{X}$ as $\frac{5}{X} + 2$ and then use

$E\left(\frac{5}{X} + 2\right) = 5E\left(\frac{1}{X}\right) + 2$ and $\text{Var}\left(\frac{5}{X} + 2\right) = 5^2 \text{Var}\left(\frac{1}{X}\right)$ to obtain answers of 9.5 and 18.75

respectively. Unfortunately many candidates could not even cope with the first step. On the other hand, there were some good attempts from evaluating

$E\left(\frac{5+2X}{X}\right)$ and $E\left(\left(\frac{5+2X}{X}\right)^2\right)$ by integration.

Q18.

Candidates seemed well able to attempt part (a) with the result that many successful outcomes were seen. Part (b)(i) was only done correctly by the most able students, with

$\int_0^t 4t(1-t^2) dt$ rarely seen otherwise. Many candidates either used 0 and 1 as their limits or simply ignored limits altogether. It was not sufficient to simply consider the indefinite integral $\int 4t(1-t^2) dt$ without consideration of the boundary conditions.

In part (b)(ii), many candidates realised that $P(\mu < T < m) = F(m) - F(\mu)$ but then went on to use the equation $2m^2 - m^4 = 0.5$ to show that $m = 0.541$ before attempting to

evaluate $\int_0^{0.541} f(t) dt$, instead of simply evaluating $0.5 - F\left(\frac{8}{15}\right)$.

Q19.

The use of calculus in questions of this type still seemed to be a popular, yet unnecessary, method for a small minority of candidates. Most were able to find the correct value of a in part (a), although it did defeat more candidates than expected. Part (b) was usually done well by those who used the correct formulae from the booklet. In part (c), many candidates failed to understand the concept of the magnitude of the error.

Q20.

In part (a), candidates showed poor ability in sketching graphs. Common faults were the omission of essential values on both axes, lines drawn as incorrectly passing through the origin on the interval $[0, 1]$ and the quadratic curve incorrectly drawn as convex on the interval $[1, 4]$. In part (b)(i), the vast majority of candidates either ignored the required limits of 0 and x altogether, or used limits of 0 and 1. Consequently, although most stated the required answer, many lost a mark. Part (b)(ii) was usually completed correctly with the correct answer of 0.85 often seen. Part (b)(iii) was not done well. Very few candidates tried to 'verify' the given result. Instead, most attempted to calculate q_x so that they could then show that its value fell within the given interval.

Q21.

In part (a), candidates showed poor ability in sketching graphs. Common faults were the omission of essential values on both axes, lines drawn as incorrectly passing through the origin on the interval $[0, 1]$ and the quadratic curve incorrectly drawn as convex on the interval $[1, 4]$. In part (b)(i), the vast majority of candidates either ignored the required limits of 0 and x altogether, or used limits of 0 and 1.

Consequently, although most stated the required answer, many lost a mark. Part (b)(ii) was usually completed correctly. Several candidates wasted time in part (b)(iii) by considering the probability density function for $x > 1$ with the inevitable loss of most, if not all, of the available marks. Those candidates who did obtain a quadratic equation usually managed to solve it correctly and nearly always realised that one of their solutions was inadmissible. Part (b)(iv) was often done well, even by those candidates who failed to appreciate what was required in part (b)(iii).

Q22.

Statistics 2A

In general, candidates seemed to have less success with this question. In part (a), the correct graph was only seen very infrequently, with the majority of candidates drawing straight lines to represent both parts of the graph. Where a curve was drawn from $t = 3$ to $t = 6$, it was often concave rather than convex. The vast majority of candidates correctly gave the answer of 'zero' to part (b). However, a minority treated this as a discrete

distribution and used $P(X = 3) = P(X \leq 3) - P(X \leq 2) = \frac{1}{5}$. Part (c) was usually done

well. However, many candidates used the more difficult integration route and, the correct answer of 0.4 was often seen. Part (d) was usually attempted quite well with most

candidates aware that the median value, m , had to satisfy $\int_0^m \frac{1}{5} dt = 0.5$. Part (e) was found difficult by all but the more able candidates. Calculating the mean to be 2.55 using the integration of two functions for the two different ranges was beyond most candidates, with several attempting to integrate one or other of the functions between 0 and 6. The final part, calculating $P(\text{median} < T < \text{mean})$ was beyond almost all but the best candidates.

Statistics 2B

In general, candidates seemed to have less success with this question than others on the paper. In part (a), the correct graph was seen very infrequently, with the majority of candidates drawing straight lines to represent both parts of the graph. Where a curve was drawn from $t = 3$ to $t = 6$, it was often concave rather than convex in nature. The great majority of candidates correctly gave the answer of 'zero' to part (b). However, a minority

treated this as a discrete distribution and used $P(X = 3) = P(X \leq 3) - P(X \leq 2) = \frac{1}{5}$. Part (c) was usually done well. However, many candidates used the more difficult integration route and, the correct answer of 0.4 was often seen. Part (d) was usually attempted quite

well with most candidates aware that the median value, m , had to satisfy $\int_0^m \frac{1}{5} dt = 0.5$. Part (e) was found difficult by all but the more able candidates. Calculating the mean to be 2.55 using the integration of two functions for the two different ranges was beyond most candidates, with several attempting to integrate one or other of the functions between 0 and 6. The final part, calculating $P(\text{median} < T < \text{mean})$ was beyond almost all but the best candidates.

Q23.

Candidates should be given more information regarding the terminology that is used in questions. Where 'state' or 'write down' are used it is to indicate to candidates that the answer can reasonably be simply written down and that working does not need to be shown. There were a great number of scripts seen where integration was used in order to find the answers to parts (a) and (b). Although the correct answer to each of these parts was eventually arrived at, it was certainly not the most efficient way of doing so. The value of $P(X > 0)$ was usually found correctly to be 0.6 but again, far too many candidates wasted time by using integration. Part (d) was only usually done correctly by the most able candidates. The vast majority of candidates usually perceived incorrectly that $P(X > 3.5)$ was required and so did not attempt to find $P(X < -3.5) + P(X > 3.5)$.

Q24.

Part (a) was generally done very well with many candidates producing a reasonably correct sketch even if by no means to scale!. Whilst there were occasional omissions, some candidates found trouble sketching the graph of $f(x) = c$, let alone the second part of the graph. Again, part (b) was done well by most candidates. Even those candidates with a deficient graph made some use of the fact that the total area was 1 unit. Sadly, in part (c), the vast majority of candidates did not convert 0.01 metres to 10 mm; were they careless or confounded? Consequently their area was totally in the rectangular section of the graph. Those candidates who considered $P(X < 10)$ invariably were successful.

Q25.

The responses to this question were very disappointing, with the great majority of candidates unable to distinguish between the distribution function, $F(x)$ as given, and the probability density function $f(x)$. Part (a) was usually only answered correctly by the most able candidates. Instead of simply evaluating $F(3) - F(2)$, the great majority integrated

$$\frac{x^3}{64}$$

before substituting values of $x = 3$ and $x = 2$. Part (b) was rarely attempted correctly

with differentiation of $F(x)$ to find the required form for $f(x) = \frac{3x^2}{64}$ infrequently seen. In

part (c)(i), most candidates assumed that $f(x) = \frac{x^3}{64}$, again revealing an inability to distinguish between the probability density function and the distribution function. In part

(c)(ii), too many candidates only attempted to work out $\int x^2 f(x) dx$, and did not then subtract $[E(X)]^2$ in order to find $\text{Var}(X)$.

Q26.

This proved a difficult question on which to score full marks. Many candidates scored full marks in part (a) though a small proportion lost marks through simple arithmetic errors. However, full marks in part (b) were rare indeed. Most of the many unsuccessful attempts centred on the incorrect use of formulae from the supplied booklet. Correct attempts usually first involved finding the value of a scaling factor so that the conditional area above 5 minutes was equal to unity.

Q27.

Sketching of the graph in part (a) was not done well. The majority of candidates incorrectly

used a straight line to represent $f(t) = \frac{6}{19}t(3-t)$ for $2 \leq t \leq 3$. Of those candidates who did attempt a curve, most incorrectly drew $f(t) = kt(t-3)$. Part (b)(i) was usually done well, with integration proving to be the most popular method. Many candidates realised that the distribution of Y , the number of times that Suneil had to wait for less than 30 seconds in order to access a telephone banking service, was given by $Y \sim B(50, 0.0395)$, with the Poisson approximation, $Po(1.975)$, correctly seen to be the required distributional approximation. Unfortunately, many candidates rounded 1.975 up to 2 and then used $Po(2)$ in Table 2 to obtain their answers. This was not correct and so gained no credit. Finding the mean in part (c), when the probability density function was given over more than one range of values, gave candidates a lot of trouble. Thus only the most able candidates managed to gain full marks for this final part of the question.

Q28.

Many candidates scored full marks on this first question. Part (a) proved to be accessible to all candidates. In part (b) a majority of candidates assumed $s = 10$, then found that the total area was $0.3 + 0.2 + 0.5 = 1$. However, about 50% of these candidates lost one mark for not stating clearly that an area of 1 is a requirement for a probability density function. Those candidates who equated an area in terms of s to 1 found the solution more difficult. However, many candidates eventually showed that, as a result, $0.05s$ must equal 0.5, or an equivalent equality.

Q29.

This question proved to be a good source of marks for many candidates. In part (a)(i),

most candidates used integration, with both $\int_6^{7.5} \left(2 - \frac{4}{15}t\right) dt$ and $1 - \int_3^6 \frac{1}{90}t^2 dt$ used and correctly evaluated. Only a few candidates, who had initially sketched the probability density function, used the area of a triangle in order to gain the given answer of 0.3.

There were many excellent attempts at part (a)(ii) with most candidates, realising that the

median value had to lie between 3 and 6, correctly used $\int_3^m \frac{1}{90}t^2 dt = 0.5$. Unfortunately, some candidates used zero as the lower limit and consequently gained a value of $m =$

6.46, this being outside the range of values of t for which the function $f(t) = \frac{1}{90}t^2$ is defined.

Part (b) was usually very well done with the vast majority of candidates realising that

$\int_3^6 t f(t) dt$ had to be considered. Most proceeded to attempt to evaluate $\int_3^6 \frac{1}{90}t^3 dt + \int_6^{7.5} \left(2t - \frac{4}{15}t^2\right) dt$. Some candidates, unfortunately, tried to evaluate one or both of these integrals using the limits 3 and 7.5.

Q30.

In part (a) the large majority of sketches scored full marks. Where marks were lost it was usually for an incorrect labelling of the vertical axis or ignoring one part of $f(x)$. The latter often resulted in a triangle in the interval 0 to 60. Correct graphs were usually followed by correct answers to parts (b) and (c). Candidates with incorrect graphs made little headway in part (b) but sometimes recovered in part (c) by calculating $1 - (\text{area of triangle from 0 to 30})$.

Q31.

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In part (a) correct diagrams having axes marked with the pertinent values were the norm.

In part (b) $\frac{1}{54}t^3$ was frequently seen but only the most able candidates presented fully correct solutions. The vast majority of candidates simply did not know how to cope with finding $F(t)$ when $f(t)$ was defined in two intervals.

Part (c) was answered very well and by a variety of methods with the majority of candidates gaining full marks.

Q32.

Almost without exception, candidates scored the first eight marks. In part (b)(iii), the conditional probability caused many candidates problems. They simply evaluated $P(X < 12)$ or quoted 0.80×0.65 . However a significant proportion of candidates, from across all

grades, scored full marks by either using $\frac{0.65}{0.80}$ directly, or by finding that, for the

(conditional) distribution between 4 and 18, $c = \frac{1}{18}$ and thus $P(x < 12) = \frac{13}{18}$. Many candidates scored the mark in part (c) usually by indicating that delays may exceed 18 minutes.

Q33.

This question was well answered with most candidates scoring most of the marks available.

In part (a), sketches were usually adequate even if a little rough at times. Some candidates lost a mark for labelling the point $2.5k$ on the vertical axis as 2.5. In part (b), correct geometrical reasoning (common) or integration (uncommon) scored full marks. Similarly, in part (c), the chosen method was invariably fully correct.

Q34.

In part (a) it was very disappointing to see $P(T = 4) = 0.25$ in a large number of scripts. The correct answer of 'zero' was seen very infrequently.

In part (b) there were many candidates who correctly considered $\int_0^{\infty} \frac{t^2}{18} dt = 0.5$ and usually went on to gain full marks. However there were many other unsuccessful attempts to show that the median was 3. These mainly consisted of the substitution of $t = 3$ into one

of the expressions $\frac{t^2}{18}$ or $\frac{1}{4}(5-t)$, giving an answer of 0.5. This then led to a false deduction.

Many very good solutions were seen to part (c) despite the indifferent starts seen in parts (a) and (b). The most common approach was by integration methods with very few candidates using the geometry of the distribution to achieve the required result. Unfortunately, a few candidates thought that <3.5 meant that the limit to use was 3.45 or 3.49.

Many excellent attempts at the correct method to find the mean were seen in part (d). Unfortunately, some candidates could not cope with the fairly simple and straightforward integration.

Q35.

This proved to be a good starter question for most candidates. In part (a), all candidates were aware that the total area had to be unity and finding this area, in terms of c , was undertaken in a variety of ways. Whilst most attempts were correct, some were confused or unclear and the solutions would have benefited from a sketch illustrating the method. Many candidates used symmetry to find the total area but few candidates equated half the area to 0.5. Similarly, very few candidates assumed $c = 0.01$ and then showed that the total area was unity. The few candidates who simply wrote $100c = 1$ so $c = 0.01$ scored at most one mark. Most candidates scored full marks in part (b). Of those who did not, most candidates used incorrect heights, particularly at $X = 20$, or found the wrong area, for example $P(X > 10)$.

Q36.

The majority of candidates showed a good understanding of this topic and so were able to score a good proportion of the marks available.

There were many very good sketches seen in part (a), although some candidates thought that $\frac{1}{54}t^2$ was the equation of a straight line. Parts (b)(i) and (ii) were well done by those candidates who used the areas of a rectangle and triangle to obtain their answers. Despite candidates being instructed to use their graphs in this part of the question, some candidates still used integration methods and so almost certainly wasted valuable time.

Part (b)(iii) was often attempted by use of $\int_0^m f(t) dt = 0.25$. Unfortunately, most candidates wrongly thought that this equated to the use of $\int_0^m \left(\frac{t^2}{54}\right) dt = 0.25$, and consequently gained no credit. The correct answer of $m = 3.5$ was rarely seen. There were many excellent solutions to part (c) with the correct answer of $\frac{4}{81}$ being seen on the majority of scripts.

Q37.

The many candidates who scored full marks for their sketches in part (a) then usually went on to score all the marks for their solutions to parts (b) and (c). Candidates whose sketches were incorrect, often through ignoring the range $0 \leq x \leq 1$, fared less well. In part

(b) they equated a triangular area of $\left(\frac{1}{2} \times 2 \times c\right)$ to 1, giving $c = 1$. In part (c), such

candidates then calculated a trapezium area of $\left(\frac{1}{2} \times \left(c + \frac{c}{2}\right) \times 1\right) = \frac{3}{4}$. These incorrect simplified solutions generally scored two out of the six marks available.

Q38.

Answers to this question were often disappointing, suggesting that candidates did not understand the rectangular distribution. In part (a), some candidates treated the distribution as discrete whilst others were apparently unaware of the formulae for μ and σ^2 in the booklet. Even those candidates who answered part (a) correctly used $\mu + \sigma^2$, rather than $\mu + \sigma$, in part (b) and then stated that $P(X \geq 13) = 0$; an answer which was awarded no marks.

Q39.

A majority of candidates scored either 7 or 8 marks on this question. In answering part (a), most candidates showed sufficient working to show they had obtained $23.5c$ (areas of rectangles, triangles, trapezia or simple counts) and equated it to unity. A proportion of candidates stated correctly the value of zero for part (b) but many stated the value of $f(250)$. Most candidates recognised the area that was required in part (c) and many evaluated it correctly, though some tried to find it as a single trapezium. A smaller number of candidates used $f(249) + \dots + f(252)$.

Q40.

Many fully correct solutions to this question were seen.

In part (a), most candidates correctly found the value of k , usually by integration. The use of the area of a triangle was surprisingly rarely seen. Since the answer was given, full credit could only be given where sufficient evidence of the correct method was seen.

In part (b), the correct form of $E(X)^2 = \int_0^4 x^2 \times \frac{1}{8} x dx$ was frequently seen. Unfortunately some candidates failed to integrate this correctly. A common incorrect response was $\frac{x^3}{3} \times \frac{1}{8} \times \frac{x^2}{2}$.

In the main, it was only the most able candidates who could cope with part (c) of this question. This part of the specification did seem to pose problems for the weaker candidates. Many candidates incorrectly assumed that $\text{Var}(5X - 4) = 25 \times \text{Var}(X) - 4$ or even $5 \times \text{Var}(X)$ instead of using $\text{Var}(aX \pm b) = a^2 \times \text{Var}(X)$.

In part (d), there were many good solutions. Most used $\int_0^a \frac{1}{8} x dx = 0.9$ to correctly calculate the 90th percentile as 3.79 correct to 3 significant figures.

Q41.

Many candidates scored full marks on this question. In part (a), there were many correct sketches though some candidates started their vertical axis at $5k$, rather than at 0: this caused them difficulties throughout the question. In answering parts (b) and (c), most candidates either used geometry or integration, though a few used a mixture. Whilst errors were rare, those candidates who used rectangles and triangles were generally less successful than those who used trapeziums and, in integration, incorrect limits were sometimes in evidence. There were few fudged answers to part (b).

Q42.

In part (a) there were many excellent solutions, with the vast majority of candidates able to manipulate fractions to obtain the correct value of k . Unfortunately some candidates used inexact decimal calculations, which was unacceptable at this level when an exact value was required. Since the answer was given in this question a full and detailed method was essential for full marks to be awarded. Although there were many excellent solutions to

part (b), several candidates wrongly thought that $P(S < 3) = \int_0^3 f(s) ds$ where $\int_1^3 f(s) ds$ was required. Those candidates who attempted to find the distribution function invariably omitted the constant of integration. The correct form of the distribution function was:

$$F(s) = \begin{cases} 0 & s < 1 \\ \frac{1}{124}(s^3 - 1) & 1 \leq s \leq 5 \\ 1 & s > 5 \end{cases}$$

This then gave $F(3) = \frac{26}{124} = \frac{13}{62} = 0.210$.

Part (c) was not well attempted, with many candidates confusing the median with the

mean and consequently evaluating $\int_1^5 sf(s)ds$ instead of using $\int_1^m f(s)ds = 0.5$, giving $m^3 = 63$ with a final answer of 4 minutes. However, some of those following the correct method lost an accuracy mark by not giving the answer correct to the nearest minute. As with part (b), some candidates thought that the lower limit was zero and consequently lost marks.

Q43.

In the main, sketches were adequate, though some candidates almost replicated the sketch given in Question 5. Adequate sketches resulted in many candidates providing a satisfactory proof in part (b) and a subsequent correct answer to part (c).

Very few candidates scored the 5 marks available in part (d). Whilst many candidates

realised that $P(2 < H < 2\frac{1}{2})$ was required and then obtained the correct value of $\frac{5}{24}$ very few then moved to the necessary conditional probability. A small number of candidates

found a new value of c , namely $\frac{1}{2}$, so that $P(2 < H < 3) = 1$ and then evaluated $P(2 < H < 2\frac{1}{2}) = \frac{5c}{4} = \frac{5}{8}$ for full marks.

Q44.

Although this was not a particularly high scoring question, candidates as a body seemed to prefer it to questions set on previous papers which have required integration. A few

candidates were unaware that $\int_1^2 (x - 1.54)^2 f(x) dx$ is the variance. Instead they expanded this integral and hence evaluated $E(X^2)$ for part (b)(iii). They then used this to evaluate the standard deviation in part (b)(ii).

Q45.

While the more mathematically able candidates had no problem with this question, others were unable to deal with the use of symbols p and q for constants.

Q46.

The proportion of candidates able to give a satisfactory answer to part (a) was much higher than that of those who could answer a similar question in January this year. Many candidates found the mean instead of the median in part (b). Most candidates picked up one or more marks in part (c), although some essentially started again instead of using the result in part (a). Only the best realised that Suggestion C was invalid as the probability density function would be negative for some values of x .

Q47.

Some candidates were thrown by this question which was in an unfamiliar form. Explaining why 0.75 was the mode caused particular problems. However, the stronger candidates scored well.

Q48.

Most candidates used integration throughout this question. Apart from part (c)(i), methods based on diagrams or quoting formulae were equally acceptable. Part (d) caused the most problems, with some apparently unsure about the meaning of 'magnitude'.



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